

ARDMS Vascular Mock Registry Exam A24

Questions first. Answer key at the end. No feedback. Corrected complete version.

1. How can a Sonographer correct for blooming on a spectral Doppler display?

- A. turn the Doppler gain down
- B. decrease the color Doppler velocity scale
- C. increase the color Doppler velocity scale
- D. reduce the wall filter

2. According to the Appropriate Use Criteria, which of the following is NOT a valid exam indication for a duplex exam on a hemodialysis fistula?

- A. palpable thrill at the fistula site
- B. ipsilateral arm swelling
- C. 1 month follow up baseline exam post-intervention
- D. Inability to cannulate at dialysis

Appropriate Use Criteria – Hemodialysis Vascular Access Duplex Ultrasound	
ALWAYS APPROPRIATE	
	• Vein mapping < 3 months before hemodialysis graft placement
	• Suspected failure to mature on clinical exam performed > 6weeks after placement
	• Graft dysfunction during dialysis
	• Inability to cannulate at dialysis (multiple personnel)
	• Palpable mass at the graft/fistula
	• Absence of palpable thrill at the graft/fistula
	• Arm swelling
	• Signs of hand ischemia (pain, pallor, ulceration)
INAPPROPRIATE	
	• Cool extremity with no other symptoms
	• Vein mapping > 3 months before hemodialysis graft placement
	• Routine surveillance

3. What native artery is commonly used for the arterial anastomosis in a renal transplant when the external iliac artery is not available?

- A. aorta
- B. external iliac vein
- C. common iliac artery
- D. internal iliac artery

4. Patients are examined in the _____ position for a lower extremity arterial Doppler exam in order to _____.

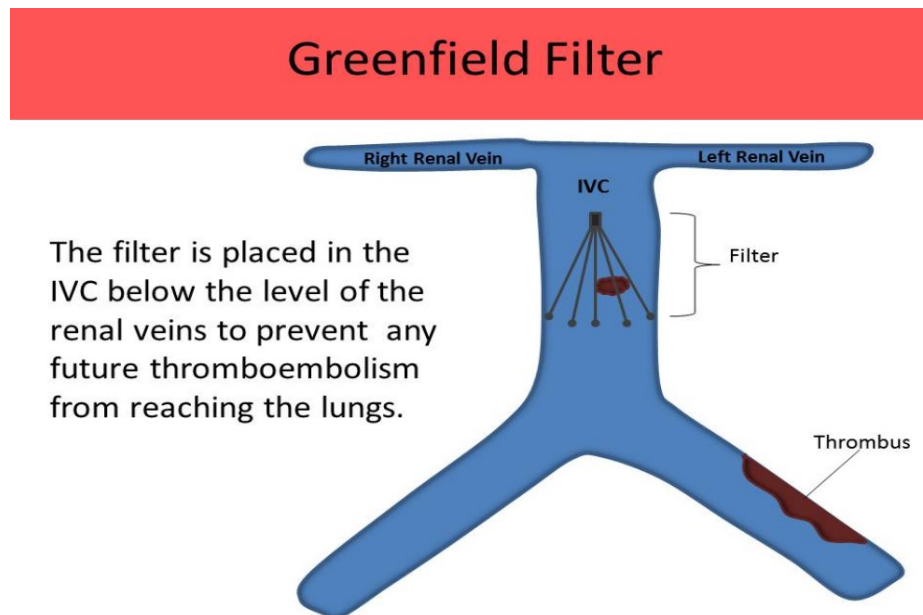
- A. semi-erect, reduce hydrostatic pressure
- B. semi-erect, increase hydrostatic pressure
- C. supine, reduce hydrostatic pressure
- D. supine, increase hydrostatic pressure

5. Which of the following can be used to describe a normal waveform obtained from a CW Doppler evaluation of the femoral artery?

- A. delayed deceleration
- B. rapid upstroke
- C. low amplitude
- D. no diastolic flow reversal

6. A patient is referred to the vascular lab to evaluate the placement of a Greenfield filter. Where will you look for this?

- A. IVC
- B. right or left groin
- C. abdominal aorta
- D. subclavian or internal jugular vein

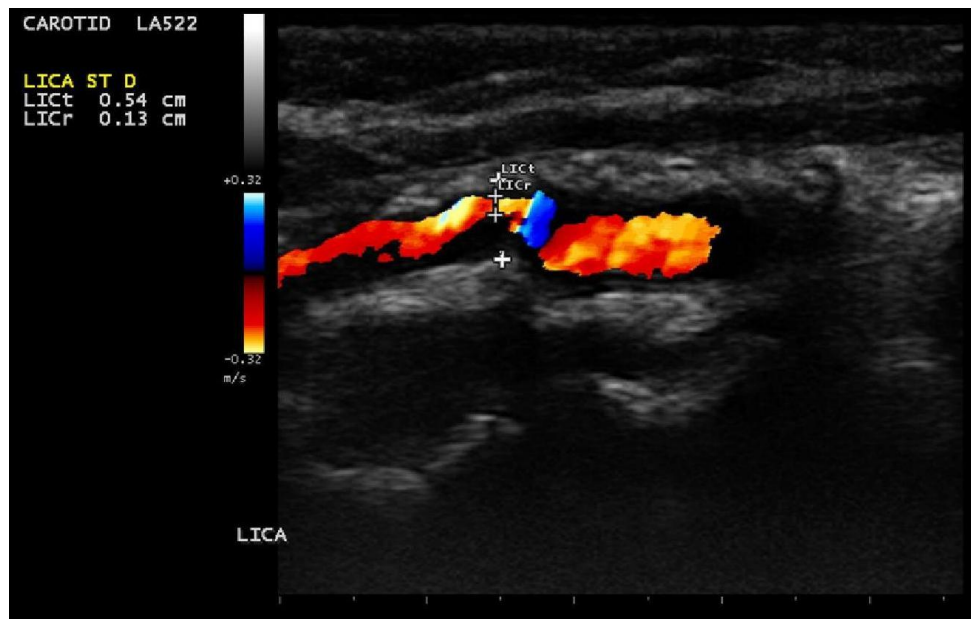


7. The transverse image of the left neck is from a carotid Doppler evaluation. What should you do next for this patient?

- A. ask the patient to perform the Valsalva maneuver
- B. evaluate the left subclavian vein
- C. attempt compression views of the neck
- D. evaluate the right subclavian vein

8. What is the diameter stenosis of the vessel on the image?

- A. 82%
- B. 24%
- C. 56%
- D. 76%



9. If a patient has a history of 40% stenosis of the left internal carotid artery identified on a MRA exam. Which of the following Doppler velocities from the ICA will correlate well with the predicted level of stenosis?

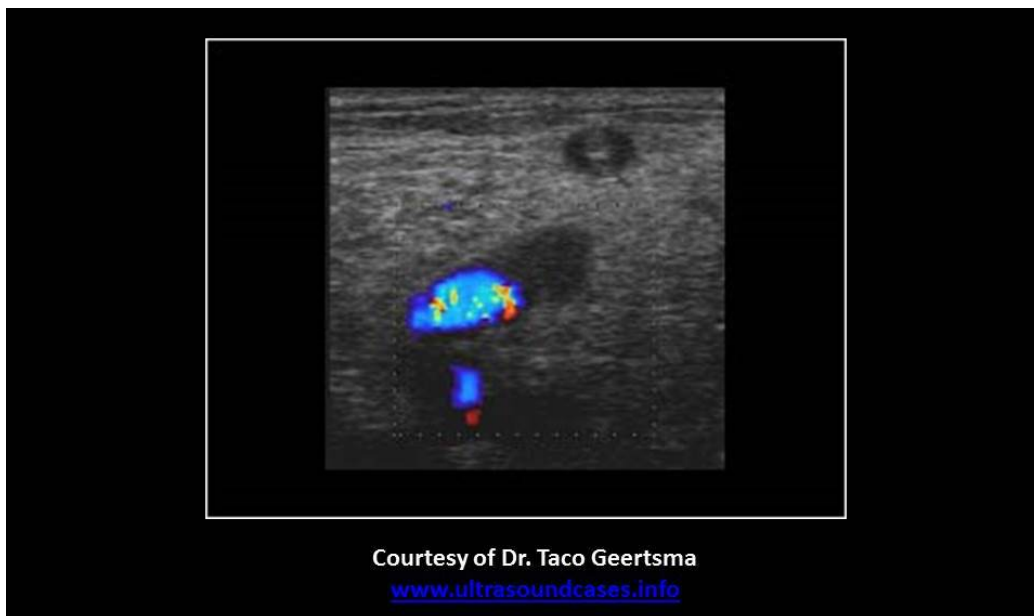
- A. PSV 120cm/s, EDV 35cm/s
- B. PSV 300cm/s, EDV 150cm/s
- C. PSV 200cm/s, EDV 100cm/s
- D. PSV 200cm/s, EDV 80cm/s

10. Thrombolytic therapy is used to treat:

- A. chronic deep vein thrombosis
- B. acute arterial occlusion caused by thrombus
- C. chronic venous disease
- D. venous insufficiency

11. The image is a transverse view of the distal popliteal fossa. Which of the following would not be a recommended treatment?

- A. IV Heparin
- B. Lovenox injection
- C. Coumadin therapy
- D. No medication, just a warm compress



12. A patient presents with a radiocephalic hemodialysis fistula in the left arm. Doppler evaluation of which of the following vessels will be used to rule out steal syndrome?

- A. Cephalic vein at the shoulder
- B. Proximal inflow artery, within 2cm above the fistula
- C. Distal inflow artery, within 2cm below the fistula
- D. Cephalic vein at the wrist

13. Which of the following procedures allows patients to continue their normal course of anticoagulant therapy before and after the procedure?

- A. ascending venography
- B. percutaneous transluminal angioplasty
- C. carotid endarterectomy
- D. compression therapy of pseudoaneurysm

14. This patient would benefit the most from what type of exam?

- A. Arterial duplex exam
- B. Aortic duplex exam
- C. Venous duplex exam
- D. Renal artery duplex exam

15. Resistive Index is a ratio utilized to quantify which of the following?

- A. phasicity of flow
- B. resistance to flow
- C. flow velocity
- D. turbulence of flow

16. A damped waveform with low velocity turbulent systolic flow will be found:

- A. at the stenosis site
- B. in a normal PTA tracing
- C. distal to a significant stenosis
- D. proximal to a significant stenosis

17. A patient presents for a bilateral segmental pressure exam of the lower extremities. The chart lists a history of a bilateral mastectomy in 1980. How will this affect your exam?

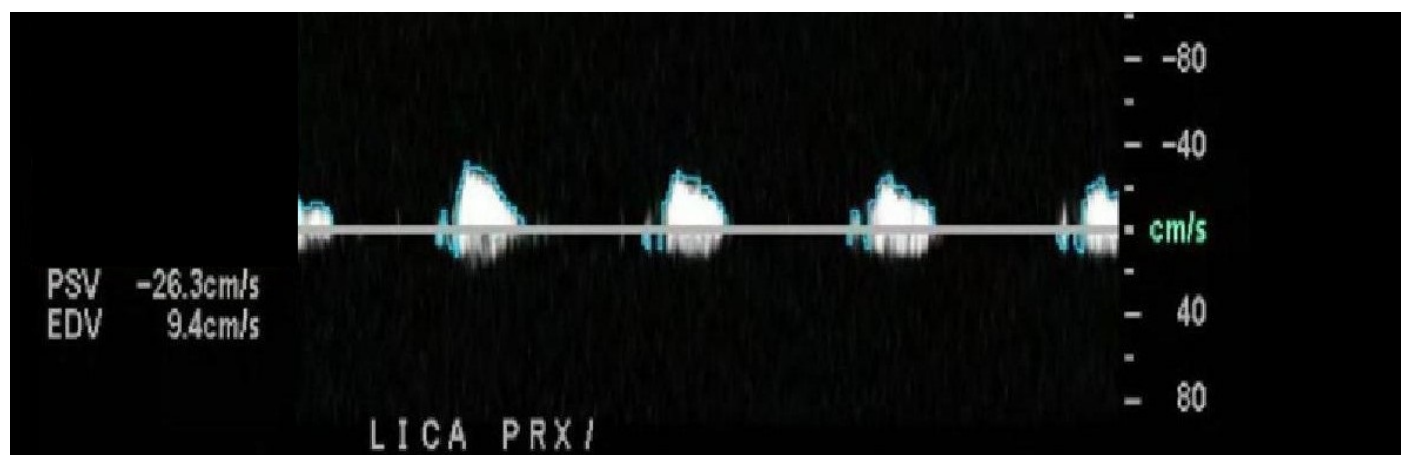
- A. The pressure indices cannot be calculated and the exam will be limited to waveform analysis only
- B. If the surgical procedure was performed more than 15 years ago, the exam should be completed using the standard protocol
- C. The patient must be evaluated in the standing position for the pressure measurements
- D. A wrist pressure must be obtained instead of a brachial pressure and this number will be used to calculate the pressure indices

18. The type of dialysis graft where the cephalic vein is anastomosed to the radial artery is called?

- A. TIPS graft
- B. synthetic loop graft
- C. Brescia-Cimino graft
- D. straight graft

19. The Doppler signal indicates:

- A. an occlusion proximal to the current point of insonation.
- B. decreased viscosity.
- C. an occlusion distal to the current point of insonation.
- D. subclavian steal on the ipsilateral side.

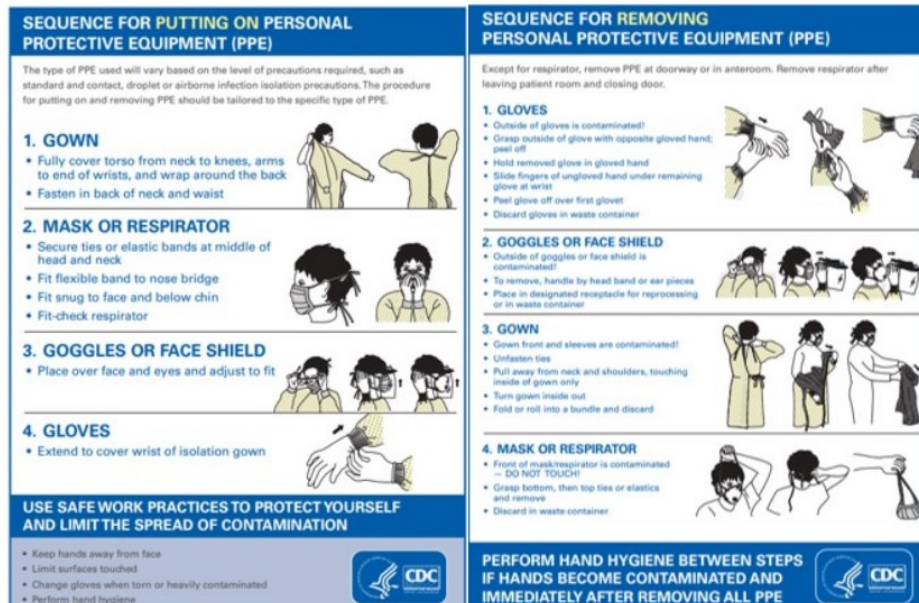


20. Which of the following will increase spectral broadening on the image?

- A. more laminar flow
- B. increased wall filter
- C. increased PRF
- D. increased sample volume size

21. What is the proper procedure for removing soiled gloves?

- A. Pull the outer cuff of the first glove toward the fingers while turning it inside out. Place the removed glove in the contralateral gloved hand and pull the outer cuff of the second glove toward the fingers while turning it inside out and enclosing the first glove.
- B. Using sterile scissors, cut the glove across the back of the hand, peel it off and dispose. Repeat the process for the second glove.
- C. Using sterile scissors, cut the glove across the palm of the hand, peel it off and dispose. Repeat the process for the second glove.
- D. Pull the first glove off using the fingertip area, while keeping the dirty side out, and dispose of it. Repeat the process for the second glove.



22. The arrow indicates what ultrasound phenomenon?

- A. Cardiac pulsatility
- B. Blood flow within the Vasa vasorum
- C. Clutter
- D. String sign

23. The most significant complication seen with DVT is _____.

- A. Varicose veins
- B. Incompetent valves
- C. Pulmonary embolism
- D. Cardiac Tamponade

24. Which of the following is most suggestive of renal transplant rejection?

- A. oliguria
- B. reduced serum levels of BUN and creatinine
- C. urinary frequency and urgency
- D. hyperbilirubinemia

25. The release of norepinephrine into the blood stream has what effect on the arterioles?

- A. decreased flow resistance
- B. stenosis
- C. vasodilation
- D. vasoconstriction

26. Which of the following terms can be used to describe the normal Doppler waveform in the portal vein?

- A. mild pulsatility with respiratory variation
- B. prominent respiratory variation with mild pulsatility
- C. blunted with respiratory variation
- D. moderately pulsatile flow due to the vessel proximity to the heart

27. According to the Bernoulli Principle, where will you find the lowest pressure in a vessel with stenosis?

- A. in the collateral vessels
- B. at the site of stenosis
- C. distal to the stenosis
- D. proximal to the stenosis

28. What technique is recommended for evaluation of the main portal vein in an obese patient with ascites and significant bowel gas?

- A. patient in the supine position with subcostal transducer position
- B. patient in the left lateral decubitus or oblique position with subcostal transducer position
- C. patient in the left lateral decubitus or oblique position with intercostal transducer position
- D. patient in the right lateral decubitus or oblique position with intercostal transducer position

29. Which of the following correctly describes a surveillance exam?

- A. testing that leads to the initial diagnosis
- B. the initial exam following vascular intervention/surgery that is usually done within first month after the procedure
- C. testing used to determine the presence or absence of disease in an asymptomatic patient
- D. testing used to monitor known disease progression

30. If a patient has a history of _____, they have an increased risk for carotid dissection.

- A. Raynaud syndrome
- B. Marfan syndrome
- C. Thoracic outlet syndrome
- D. Budd Chiari

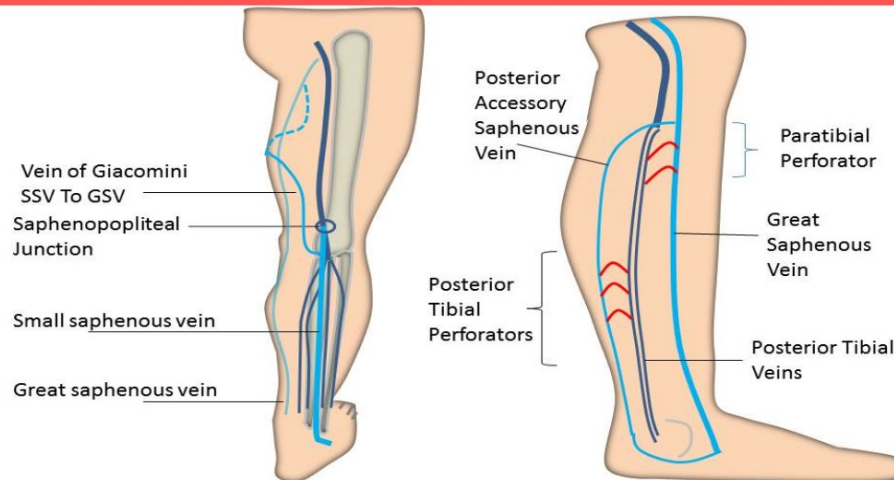
31. Which of the following are branches of the distal ICA?

- A. supraorbital artery and posterior cerebral artery
- B. middle, anterior and posterior cerebral artery
- C. middle cerebral artery, anterior cerebral artery and posterior communicating artery
- D. middle cerebral artery and anterior communicating artery

32. _____ receives the blood from the posterior tibial perforators and plays a major role in venous stasis of the lower extremity.

- A. The anterior accessory saphenous vein
- B. The vein of Giacomini
- C. The lateral perforator
- D. The posterior venous arch

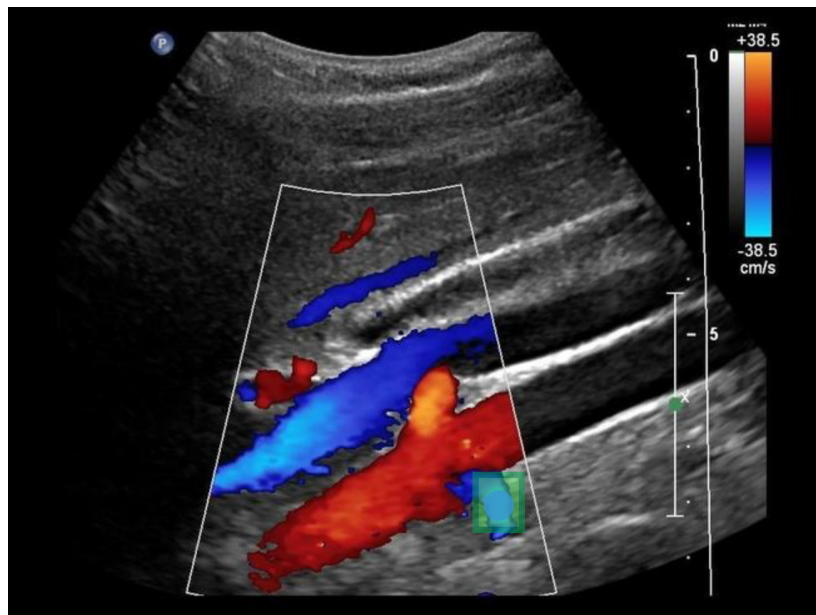
Communicating Veins



33. The mid left femoral vein appears to be occluded by hypoechoic, non-compressible thrombus. The popliteal vein is completely compressible. Which of the following Doppler waveform characteristics is expected from the popliteal vein?

- A. decreased response to distal augmentation of calf muscle
- B. normal response to distal augmentation of calf muscle
- C. loss of respiratory phasicity
- D. increased pulsatility

34. Use your mouse to place your cursor over the left renal artery and click to mark the vessel.



35. According to the findings on this arteriogram, which of the following symptoms would be explained by the findings?

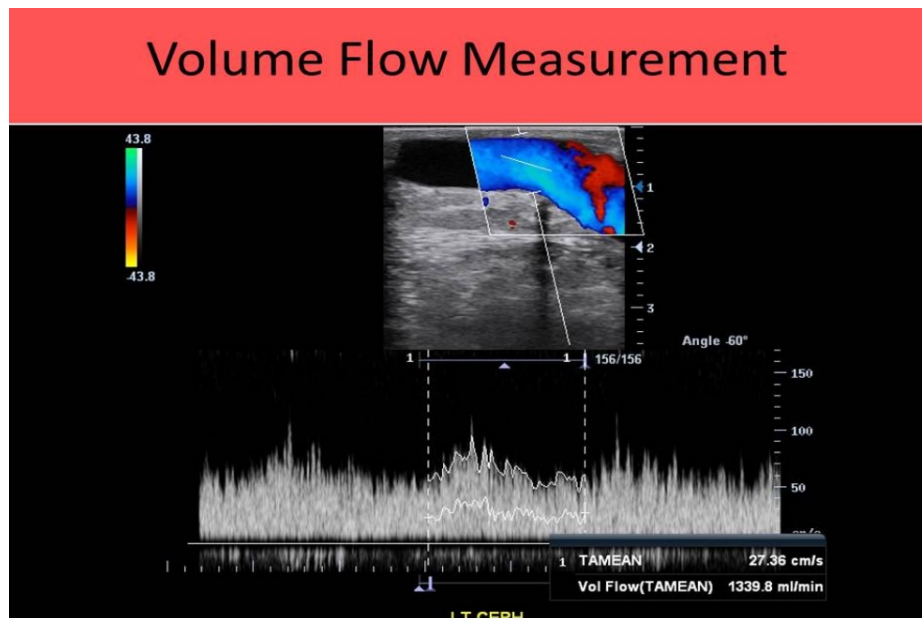
- A. Left arm paresthesia
- B. Right arm paralysis
- C. Left leg paresthesia

D. Isolated visual disturbances in the left eye



36. Volume flow calculations are used to assess hemodialysis graft performance. How is volume flow measured?

- A. Open the sample volume from wall to wall within a normal segment of the graft and trace the waveform.
- B. Obtain 3-4 Doppler tracings of the inflow artery and trace the waveforms to obtain the volume flow measurements, then average these numbers.
- C. Obtain 3-4 Doppler tracings and average the peak velocity, then use the Bernoulli equation to calculate the volume flow.
- D. Doppler US cannot provide the volume flow information, and the dialysis machine must be used to assess the volume flow through the graft.



37. Which of the following measurements should be obtained to evaluate the success of a percutaneous angioplasty performed on the left proximal superficial femoral artery?

- A. acceleration time of the distal superficial femoral artery
- B. femoral artery diameters proximal to the treated area, the treated area and distal to the treated area

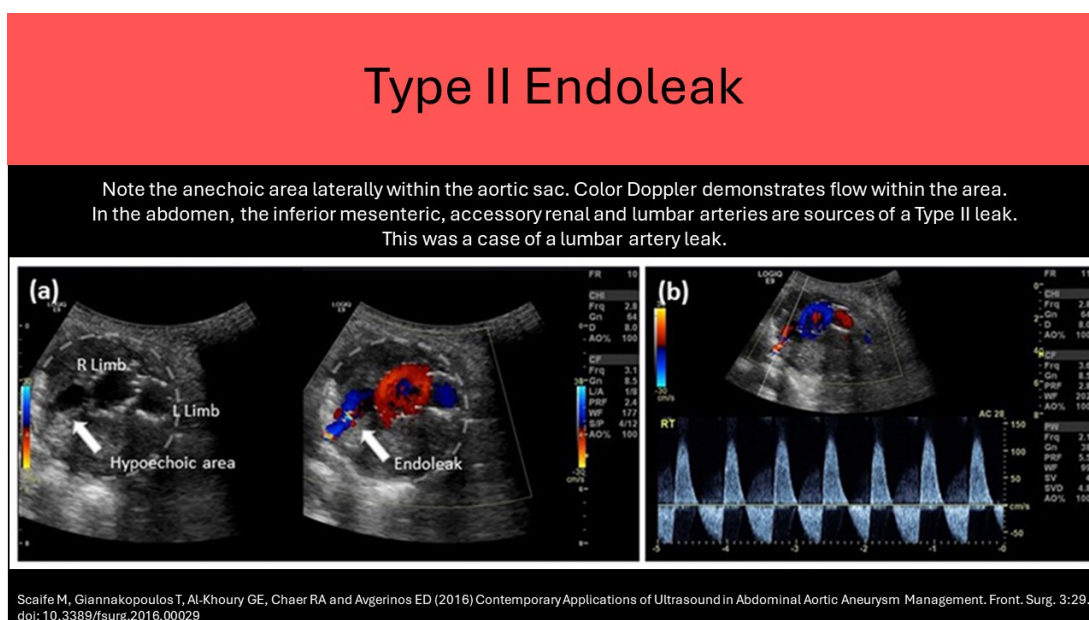
- C. evaluate the ABI value for the expected decrease
- D. obtain waveforms at the PTLA site and measure the peak systolic velocity and pulsatility index

38. A patient presents for a follow up exam after a recent endarterectomy that was performed 3 days ago. The intraoperative report indicates a possible shelf lesion in the ICA. How will this be demonstrated on today's exam?

- A. <3mm intimal flap that causes turbulence and increased velocity
- B. >3mm intimal flap that causes turbulence and increased velocity
- C. <3mm intimal flap demonstrates a false lumen with turbulent flow
- D. Hyperechoic plaque in the vessel lumen with an abrupt edge

39. Which of the following describes the most common flow pattern seen in a Type II endoleak?

- A. monophasic, low resistance
- B. bidirectional
- C. continuous, phasic flow
- D. monophasic retrograde flow



40. Lower extremity arterial segmental pressure exam results in a post exercise basal pressure recovery time of > 6 minutes which suggests:

- A. Popliteal entrapment syndrome
- B. Single level stenosis
- C. Iliac disease
- D. Multi-level stenosis

41. Thoracic outlet syndrome occurs when the _____ is compressed by the _____.

- A. superior mesenteric artery, median arcuate ligament of the diaphragm
- B. renal artery, diaphragm crura
- C. brachial plexus, scalene muscle or cervical rib
- D. celiac axis, median arcuate ligament of the diaphragm

42. Which of the following patients has the lowest risk of upper extremity venous thrombosis?

- A. recent shoulder surgery

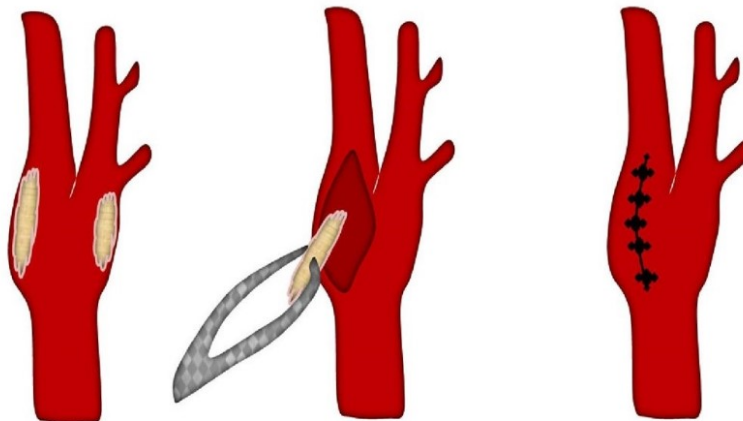
- B. hemophilia
- C. central venous catheter for chemotherapy
- D. polycythemia vera

43. _____ refers to when the carotid artery is lacerated and the atheroma formation is scraped for removal.

- A. An angioplasty procedure
- B. An atherectomy procedure
- C. An endarterectomy procedure
- D. A valvulotome procedure

Endarterectomy

During an endarterectomy, the carotid artery is lacerated and the atheroma formation is scraped for removal.



44. Proximal augmentation during an extremity Doppler evaluation is used to assess:

- A. vein compressibility
- B. arterial insufficiency
- C. venous insufficiency
- D. for the presence of CHF

45. A patient presents for a renal transplant evaluation. Which of the following native vessels will be evaluated during the exam?

- A. segmental artery
- B. distal renal artery
- C. external iliac artery and vein
- D. arcuate arteries

46. If new research shows that the minimum velocity criteria used to diagnose $>60\%$ renal artery stenosis is actually 225cm/s , not 180cm/s , how will this affect the negative predictive value of the exam?

- A. decrease will vary with patient population
- B. no effect on the negative predictive value, only affects accuracy
- C. decrease by 4.5%
- D. increase by 4.5%

47. Thromboangiitis obliterans involves inflammation of:

- A. intimal wall layer
- B. all vessel walls with no effect on connective tissue
- C. intimal and media wall
- D. all vessel walls and surrounding connective tissue

48. When compared to the normal left CCA, the right CCA demonstrates reduced velocity, slower upstroke and holodiastolic flow. These findings are most consistent with:

- A. stenosis proximal to the point of insonation
- B. decreased hematocrit levels
- C. elevated hematocrit levels
- D. congestive heart failure

49. A patient presents with leg pain 2 days after a coronary artery catheterization. All of the following are an important part of the physical evaluation of the patient, except :

- A. Evaluation of the leg for signs of pallor
- B. Evaluation of the lower leg for loss of hair, dry skin and thick toenails
- C. Evaluation of the toes for blue discoloration
- D. Evaluation of the groin for a bruit

50. Venous PPG tracings are obtained. The right side demonstrates a venous refill time of 14 seconds. The left side demonstrates a refill time of 23 seconds. Which of the following describes the next step in evaluation?

- A. The results are normal. No further evaluation is necessary.
- B. A tourniquet will be applied to the left leg at the thigh level and the VRT assessment will be repeated.
- C. A tourniquet will be applied to both legs at the thigh level and the VRT assessment will be repeated.
- D. A tourniquet will be applied to the right leg at the thigh level and the VRT assessment will be repeated.

51. Which of the following is not an expected complication in a 3-week post-op liver transplant patient?

- A. portal vein thrombosis
- B. hepatic vein thrombosis
- C. hepatic artery atherosclerosis
- D. biloma

52. Which of the following veins normally do not demonstrate spontaneous flow on Doppler evaluation?

- A. posterior tibial and popliteal veins
- B. peroneal and femoral veins
- C. posterior tibial, peroneal and great saphenous vein
- D. All lower extremity veins should normally demonstrate spontaneous flow on Doppler evaluation. Loss of spontaneity indicates flow stasis.

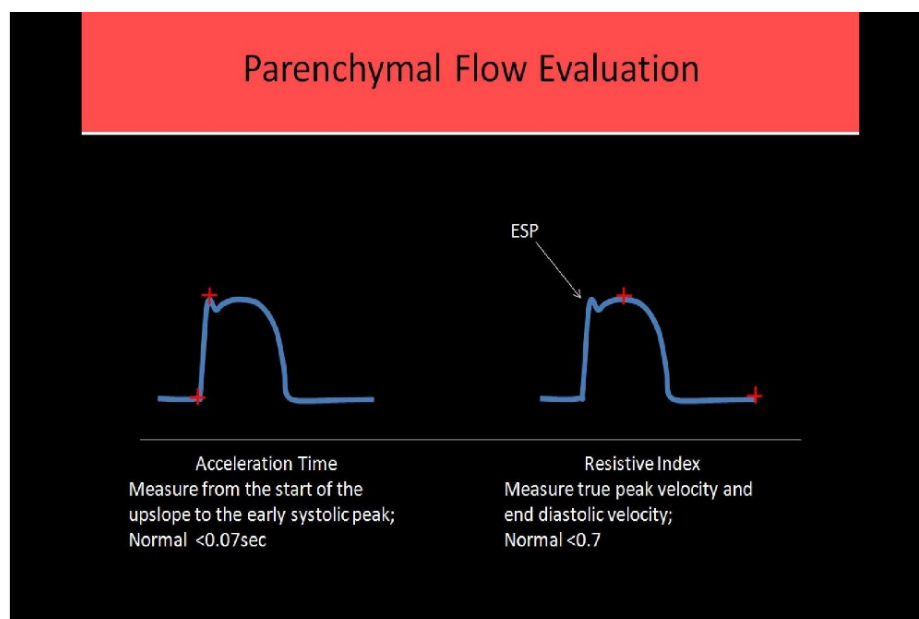
53. Abdominal Doppler exams are usually best performed with a frequency of _____MHz and Lower Extremity Arterial Duplex exams are usually best performed with a frequency of _____MHz.

- A. 5-7MHz, 8-10MHz
- B. 2-4MHz, 8-10MHz
- C. 1-3MHz, 5-7MHz
- D. 2-4MHz, 5-7MHz

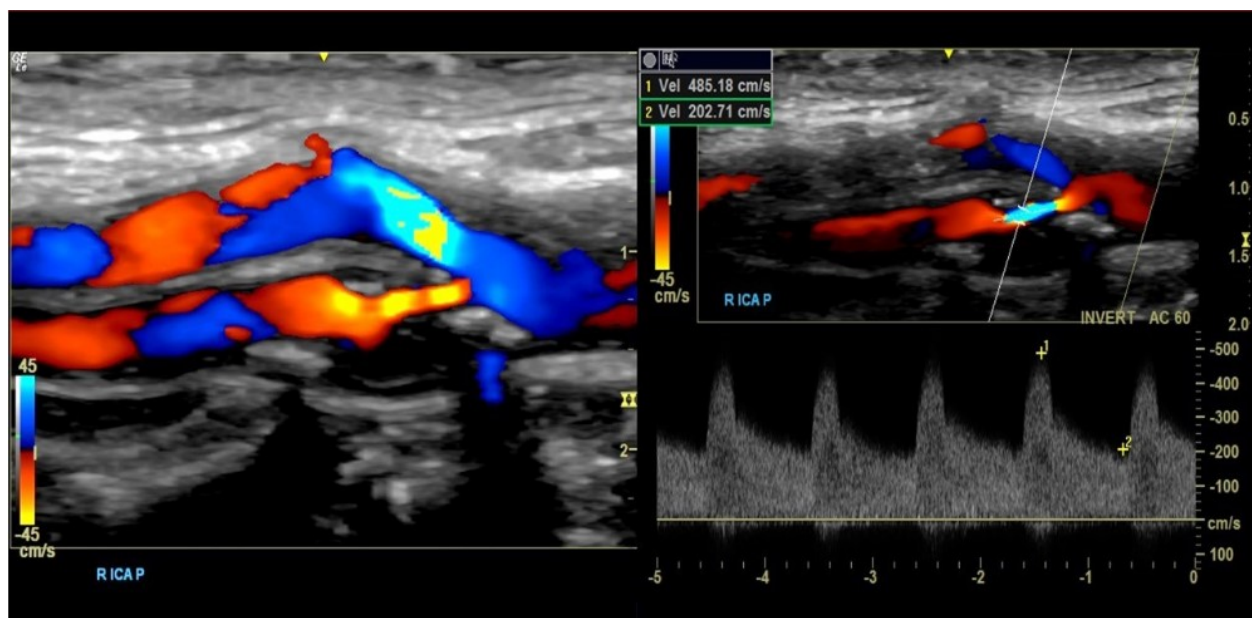
54. Which of the following describes the correct way to measure the acceleration time for a renal artery waveform?

- A. place a caliper at the early systolic peak and the end diastolic velocity
- B. place a caliper at the onset of the systolic upstroke and at the true systolic peak

- C. place a caliper at the onset of the systolic upstroke and at the early systolic peak
- D. place a caliper at the true systolic peak and the end diastolic velocity



- 55. A peak systolic velocity of 100 cm/s is an indication of _____ in the ICA.**
- A. normal flow
 - B. mild stenosis
 - C. moderate stenosis
 - D. severe stenosis
- 56. Which of the following normally demonstrates a low resistance Doppler flow profile?**
- A. external carotid artery
 - B. hepatic artery and renal artery
 - C. renal artery and internal iliac artery
 - D. external carotid artery, renal and hepatic arteries
- 57. When evaluating the Doppler waveform at a focal stenosis in the proximal ICA, which of the following Doppler characteristics will be demonstrated?**
- A. increased peak systolic velocity, increased end diastolic velocity
 - B. decreased peak systolic velocity, increased end diastolic velocity
 - C. increased peak systolic velocity, decreased end diastolic velocity
 - D. increased peak systolic velocity, increased diastolic flow reversal



58. Which of the following is a contraindication for a lower extremity segmental pressure exam?

- A. diabetes mellitus
- B. current anticoagulant treatment
- C. ulceration on the 3rd and 4th toes
- D. ulceration on the distal calf

59. Which of the following conditions would benefit from intravascular ultrasound with repair?

- A. cerebral artery aneurysm only
- B. AAA and renal artery stenosis
- C. renal artery stenosis only
- D. AAA only

60. If an abnormal transcutaneous oximetry reading is obtained, what should you do next?

- A. report the reading location and end the exam
- B. turn up the oxygen level entering the patient's mask
- C. move the sensor distally
- D. move the sensor proximally

61. Paresthesia is defined as:

- A. loss of vision in one eye for a short period of time
- B. loss of ability to move an extremity or other body part
- C. dizziness related to change in patient position
- D. an abnormal skin sensation, such as numbness or tingling

62. Which imaging plane can demonstrate the IVC and the aorta on the same image in a normal patient?

- A. coronal only
- B. mid sagittal only
- C. transverse only
- D. coronal and transverse

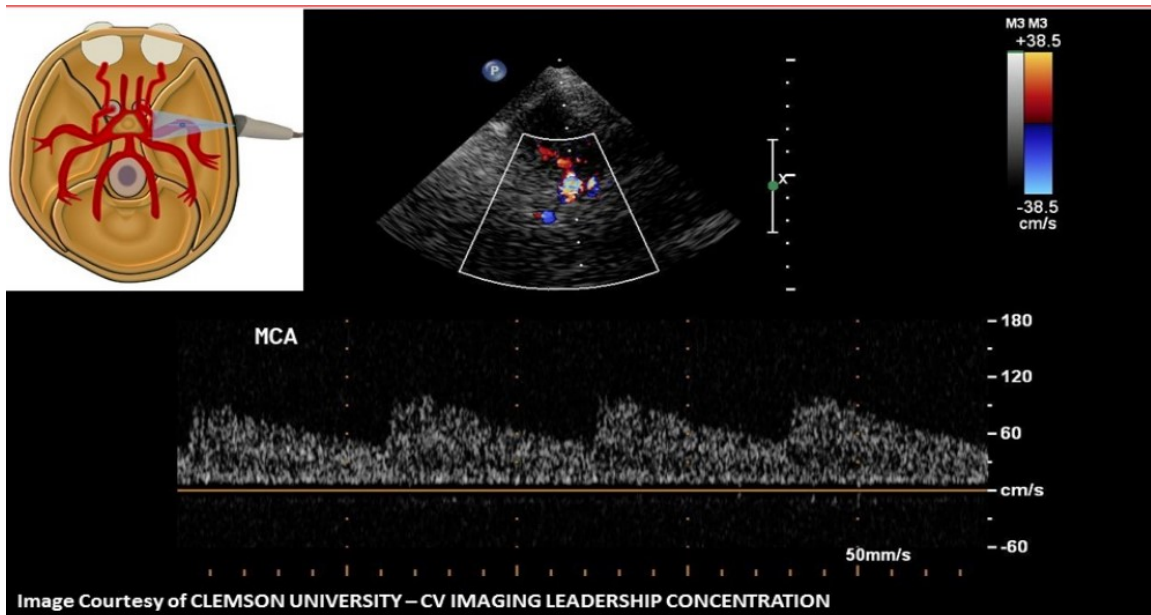
63. When scanning a patient with a history of renal cell carcinoma, you identify a hypoechoic mass within the IVC lumen near the renal veins. What is the best way to determine if this is tumor invasion or

thrombus formation?

- A. Roll the patient into the decubitus position to evaluate the possible mobility of the mass.
- B. Evaluate the iliac veins for thrombosis.
- C. Apply color Doppler to assess internal vascularity of the mass.
- D. Evaluate the aorta for tumor invasion.

64. Which of the following describes the normal appearance of a waveform obtained in the middle cerebral artery?

- A. low resistance, laminar flow with clear spectral window
- B. low resistance flow that varies with patient respiration
- C. low resistance with spectral broadening
- D. low resistance, low velocity with small flow reversal component in diastole



65. Which of the following can be found immediately posterior to the inguinal ligament?

- A. profunda femoral vein
- B. common femoral vein and artery
- C. popliteal vein and artery
- D. saphenopopliteal junction

66. Which of the following veins does not normally contain any functioning valves?

- A. Innominate vein
- B. Axillary vein
- C. Peroneal veins
- D. Small saphenous

Lower Extremity Veins No Valves	Lower Extremity Veins With Valves	Upper Extremity Veins No Valves	Upper Extremity Veins With Valves
IVC	Common Femoral Vein (1 valve)	SVC	Jugular Veins
Internal Iliac Vein	Popliteal Veins (1-3 valves)	Innominate Veins	Subclavian Veins
Common Iliac Vein	PTV, ATV, PER Veins (≤12 valves)		Other UE veins may contain valves
External Iliac Vein (75% patients)	Short Saphenous Veins (6-12 valves)		
Soleal Sinuses	Great Saphenous Veins (12-14 valves, most located below the knee)		

67. Which of the following arteries is normally only an intracranial vessel?

- A. vertebral artery
- B. internal carotid artery
- C. superficial temporal artery
- D. basilar artery

68. According to the Appropriate Use Criteria, which of the following is an appropriate indication for a lower extremity venous duplex exam?

- A. Screening after recent orthopedic surgery
- B. Suspected or diagnosed pulmonary embolism
- C. Positive D-dimer test
- D. > 4 days in the intensive care unit

69. Which of the following describes the probe placement for a Doppler evaluation of the frontal artery?

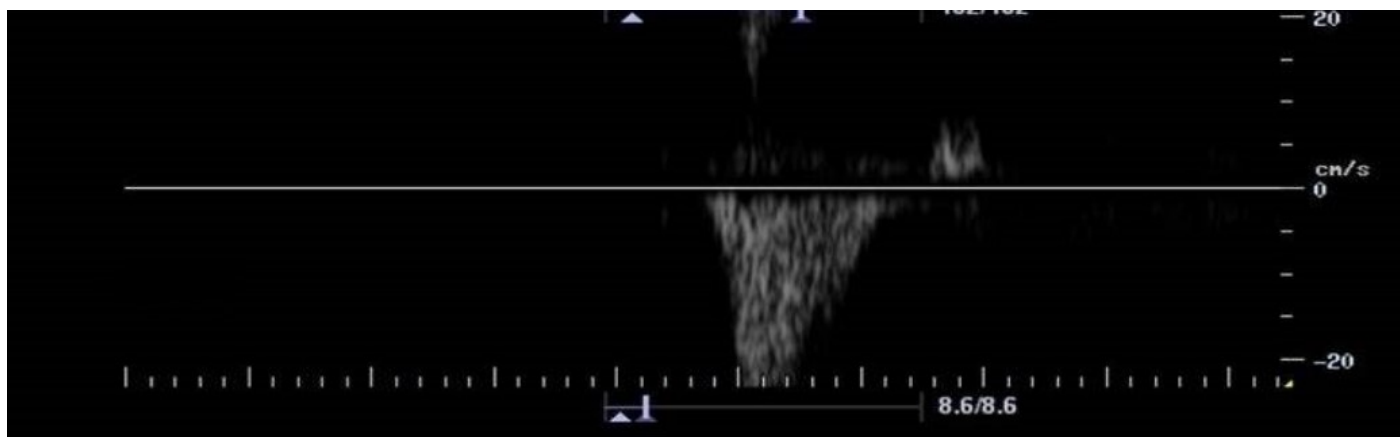
- A. near the inner canthus of the eye
- B. near the outer canthus of the eye
- C. just above the mid-point of the eyebrow
- D. approximately 2cm above the zygomatic arch

70. A patient presents for penile Doppler and segmental pressure exam due to impotence. The ABI is >1.0 bilaterally. The PBI is 0.8. What should be done next to evaluate this patient?

- A. Use papaverine to assess for increased venous outflow
- B. Write the results as normal with no explanation of impotence
- C. Use papaverine to assess for decreased venous outflow
- D. Have the patient perform 5 minutes of exercise on the treadmill at 12% grade

71. The attached image is taken from a lower extremity venous Doppler exam. This flow pattern would be abnormal for which of the following veins?

- A. posterior tibial veins
- B. anterior tibial veins
- C. great saphenous vein
- D. popliteal vein



72. Which of the following is true regarding compression techniques used to treat a pseudoaneurysm of the femoral artery in the groin?

- A. Pressure equivalent to 100mmHg should be performed in three ten-minute intervals and if this does not close the stalk, alternative treatment must be considered
- B. Thrombin injection is more effective than compression techniques in pseudoaneurysms with large diameter stalks.
- C. Compression should be performed in 30 second intervals with color Doppler evaluation in between compressions.
- D. Blood pressure evaluation cannot be performed anywhere on the legs until the stalk to the pseudoaneurysm is closed.

73. When you reduce angle correction to make the incident beam more parallel to the blood flow, how does the peak velocity displayed change?

- A. no change unless stenosis is present
- B. increase
- C. decrease
- D. no change unless the patient is hypertensive

74. In most cases, a peripherally inserted central catheter (PICC) line is inserted in the basilic vein exam and advanced until the tip reaches:

- A. brachiocephalic vein
- B. atriocaval junction
- C. proximal subclavian vein
- D. axillary vein

75. If the velocities in both carotid arteries are increased since last year's exam with no visible change in plaque formation, what should you review in the patient chart as a possible explanation?

- A. medical history for controlled HTN
- B. medical history for polycythemia vera
- C. lab values for decreased hematocrit levels
- D. recent echocardiogram for signs of decreased cardiac output

76. A Greenfield filter is used to prevent which of the following?

- A. Cardiac tamponade
- B. DVT
- C. Pulmonary embolism
- D. Blue toe syndrome

77. If a liver transplant patient has an interposition anastomosis, how does this affect your evaluation?

- A. each hepatic vein is connected to the native IVC separately, so there will be three anastomosis sites in the IVC that must be evaluated

- B. there will be two anastomosis sites in the main portal vein that must be evaluated
- C. there will be two anastomosis sites in the IVC that must be evaluated
- D. the hepatic artery anastomosis will not be able to be visualized because it is located deep within the liver tissue

78. Where does the first caliper go when measuring the acceleration time on PW Doppler? Use your mouse to position the cursor over the correct part of the 3rd waveform and click to set your answer.



79. Which of the following Doppler angles would lead to the most significant error in estimation of flow velocity?

- A. 25
- B. 45
- C. 65
- D. 80

80. A patient is scheduled for a stab phlebectomy. What is being treated for this patient?

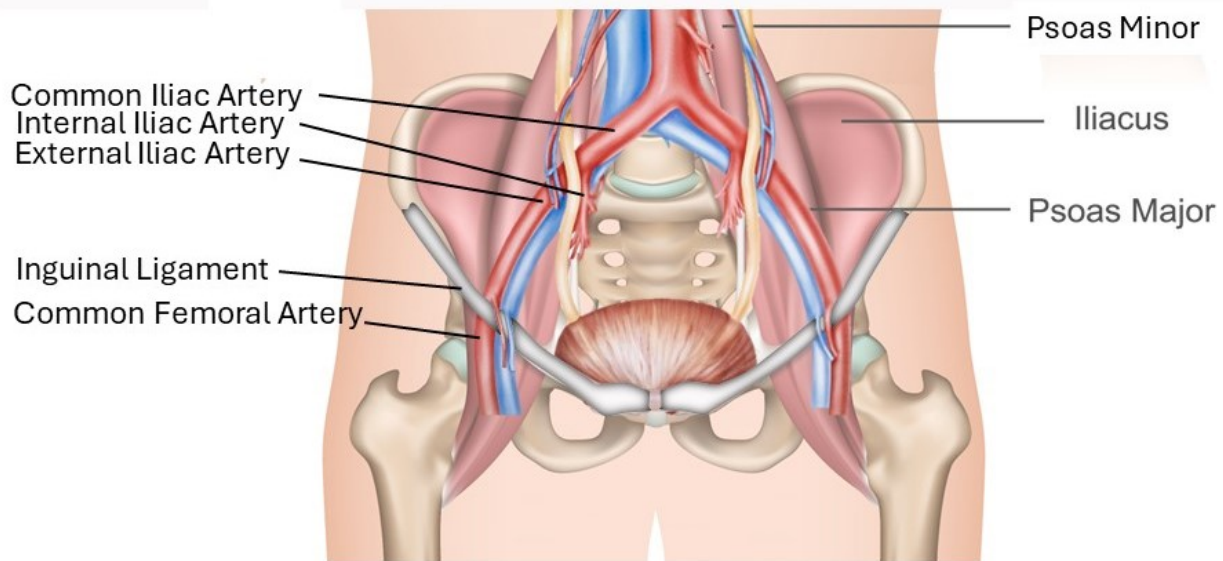
- A. DVT removal
- B. pseudoaneurysm
- C. carotid stenosis
- D. varicose veins

81. Which of the following is considered an abnormality that causes turbulence in a vessel?

- A. bifurcation
- B. tortuosity
- C. eccentric change in vessel course
- D. myointimal hyperplasia

82. The inguinal ligament:

- A. connects the superior edge of the iliac crest with the anterior pubic bone
- B. connects the ischial tuberosity to the anterior pubic bone
- C. connects the greater and lesser tuberosities of the femur
- D. connects the greater tuberosity to the superior edge of the iliac crest

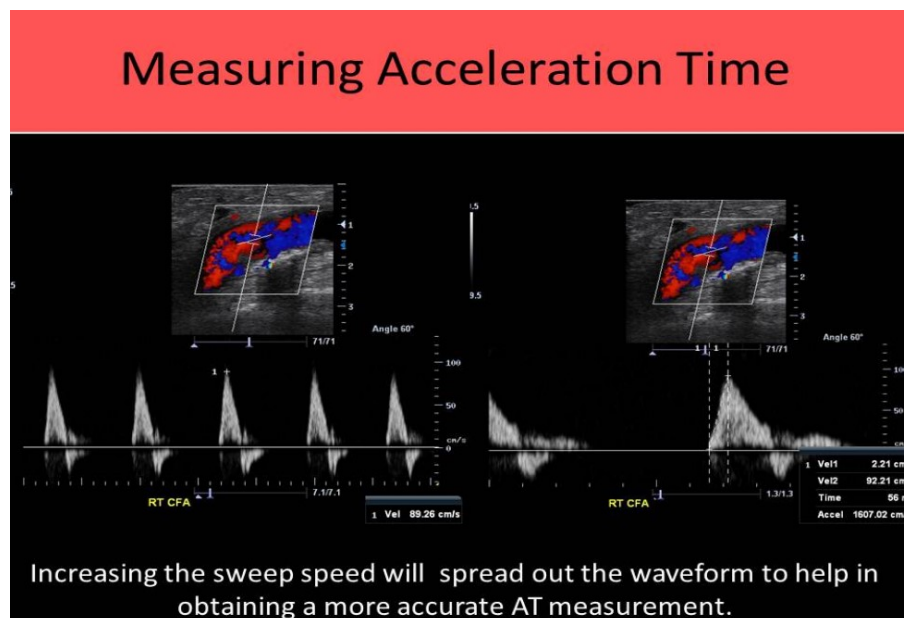


83. Which of the following is an expected clinical finding in a patient with a venous stasis ulcer?

- A. deep crater in the skin
- B. severe pain
- C. dry and crusty
- D. located at the medial malleolus

84. _____ aids in measurement of the acceleration time of the renal arteries.

- A. Using a lower frequency transducer
- B. Increasing the sweep speed
- C. Increasing the PRF
- D. Reducing rejection



85. Which of the following is a potential collateral pathway for extracranial to intracranial blood flow?

- A. Ascending pharyngeal branches connect to small branches of the lingual artery
- B. Superficial temporal artery branches connect to small branches of the ascending pharyngeal artery

- C. Ascending pharyngeal branches connect to small branches of the superior thyroidal artery
- D. Facial and maxillary artery branches connect to small branches of the orbital artery

86. Which of the following correctly describes the color displayed on the image?

- A. The heel-to-toe method can be used to change the course of the vessel across the image, and this will allow the vessel to accurately display a single color
- B. The color scale should be reduced so that the entire vessel is filled with red color.
- C. The red and blue colors indicate that flow is moving in two different directions simultaneously in the vessel
- D. The color scale should be reduced so that the entire vessel is filled with blue color.

87. If an arteriogram of the carotid arteries indicates a 60% stenosis of the left internal carotid artery, which of the following Doppler velocities from the ICA will correlate well with the predicted level of stenosis?

- A. PSV 125cm/s, EDV 40cm/s
- B. PSV 300cm/s, EDV 150cm/s
- C. PSV 200cm/s, EDV 80cm/s
- D. PSV 200cm/s, EDV 160cm/s

88. Which of the following describes chronic lower extremity DVT?

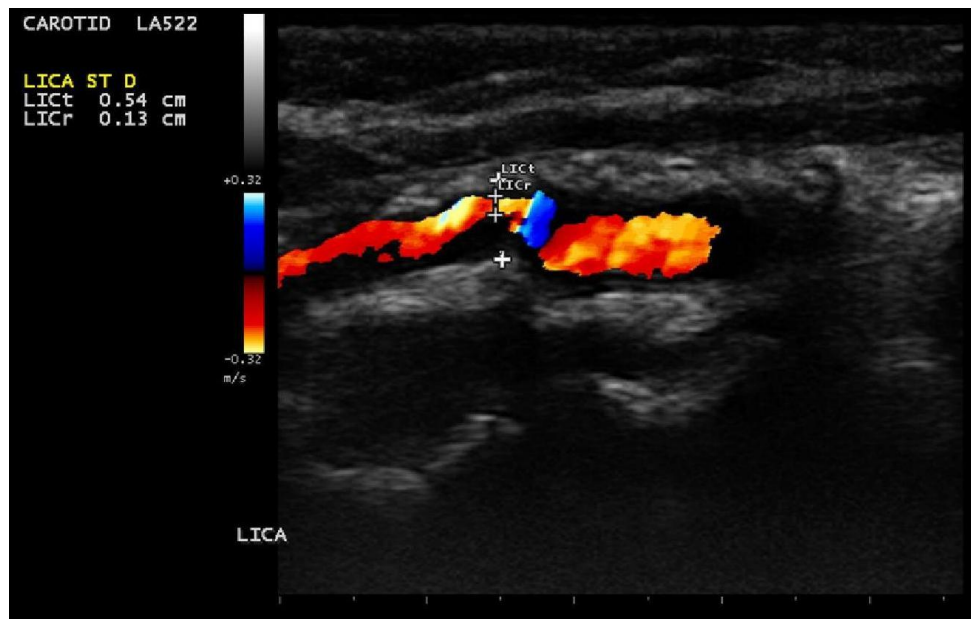
- A. vein is completely non-compressible
- B. echogenic vein walls
- C. exaggerated respiratory phasicity
- D. pulsatile flow pattern

89. Which of the following is not a branch of the subclavian artery?

- A. thyrocervical trunk
- B. posterior auricular artery
- C. costocervical trunk
- D. internal thoracic artery

90. What is the expected waveform characteristic if the Doppler sample volume is placed at the same location as the calipers?

- A. triphasic with increased peak systolic and end diastolic velocities
- B. blunted waveform with increased diastolic flow
- C. increased peak systolic velocity and decreased end diastolic velocity
- D. monophasic with increased peak systolic and end diastolic velocities



91. A diabetic patient presents with an injury to his big toe that won't seem to heal and has been open for over 3 weeks. Which of the following exam techniques should be used to evaluate this patient?

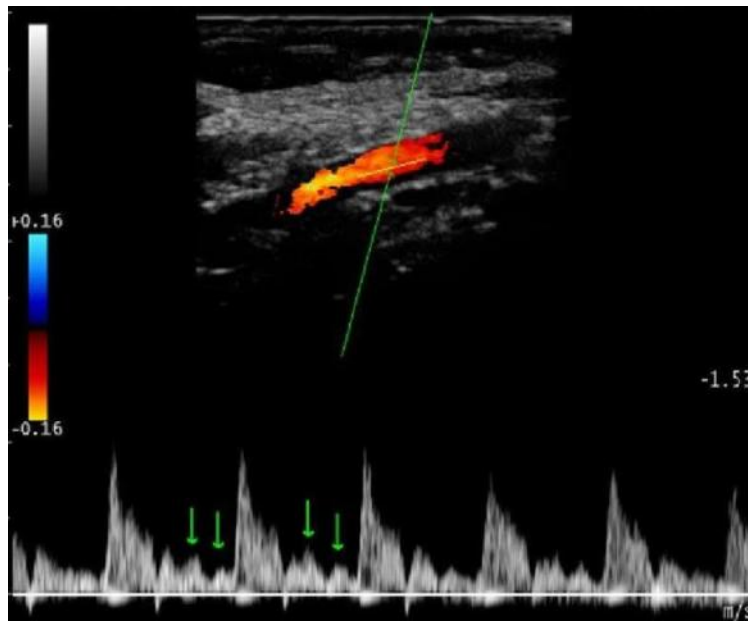
- A. segmental pressure exam
- B. digital pressures and PPG evaluation
- C. cold sensitivity testing
- D. treadmill testing

92. What is the most common cardiac disorder associated with arterial embolism?

- A. cardiomyopathy
- B. atrial fibrillation
- C. myocardial infarction
- D. Marfan syndrome

93. What is occurring on the Doppler tracing where the arrows are pointing?

- A. Patient is swallowing
- B. Patient is snoring
- C. Temporal tap
- D. Atrial fibrillation



94. When evaluating the GSV for use as a bypass graft, what is the preferred diameter for the vein?

- A. >7mm
- B. >3mm
- C. >10mm
- D. >5mm

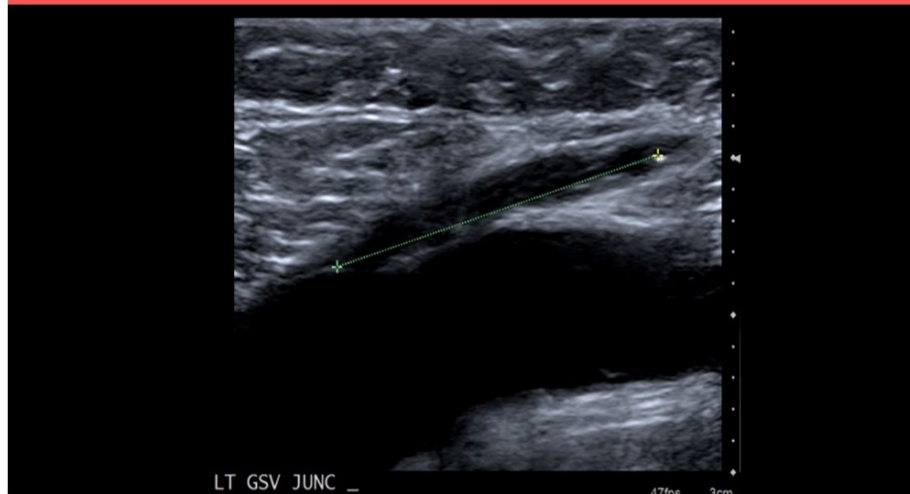
95. This image demonstrates which of the following?

- A. Chronic DVT
- B. Partial DVT
- C. Acute DVT
- D. Superficial venous thrombosis

96. A patient presents for a follow up exam after radiofrequency ablation of the GSV. What is the purpose of the exam?

- A. To assess the valvular motion in the GSV
- B. To evaluate thrombus formation in the GSV
- C. To confirm the patency of the GSV and thrombosis of its branches
- D. To confirm the contraction of the GSV and absence of thrombus in the deep system

Post-Ablation Assessment



97. Which of the following is least likely to be associated with renal artery stenosis?

- A. Fibromuscular dysplasia
- B. May Thurner syndrome
- C. Scleroderma
- D. CREST syndrome

98. Which of the following demonstrates the correct calculation for the Lindegaard Ratio?

- A. MCA mean flow velocity divided by distal ICA mean flow velocity
- B. siphon peak systolic velocity divided by proximal ICA peak systolic velocity
- C. ACA peak systolic velocity divided by distal ICA peak systolic velocity
- D. MCA mean flow velocity divided by siphon peak systolic velocity

99. An AV fistula is present connecting the femoral artery and vein at the right groin. The Doppler evaluation of the common femoral artery will demonstrate what flow pattern?

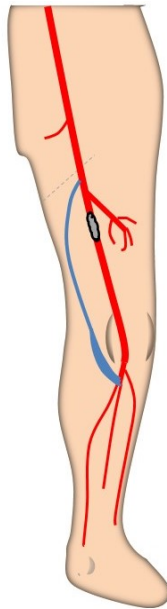
- A. monophasic with increased diastolic flow
- B. peaked, triphasic waveform
- C. biphasic flow with increased diastolic flow reversal
- D. monophasic with minimal diastolic flow

100. The waveform from an abdominal vessel displays what two characteristics?

- A. triphasic flow and no respiratory variation
- B. mild phasicity and pulsatility
- C. flow reversal during diastole
- D. flow reversal during augmentation

101. What type of venous graft is demonstrated in the diagram?

- A. In vivo
- B. In situ
- C. Reversed
- D. Greenfield



102. What blood vessels are located at letter B?

- A. 1 ATA and 2 ATV
- B. 2 peroneal arteries and 1 peroneal vein
- C. 1 PTA and 2 PTV
- D. 2 peroneal veins and 1 peroneal artery

103. The common femoral artery velocity is 0.8 m/s and the proximal femoral artery demonstrates atherosclerosis formation causing an increase in the velocity to 2.2 m/s. This indicates what level of stenosis?

- A. 25%
- B. 50%
- C. 75%
- D. 80-99%

104. Which of the following statements is true regarding the aorta?

- A. Rupture is the most common complication of an aortic aneurysm.
- B. Suprarenal AAAs normally do not affect the arterial flow to the kidneys.
- C. The majority of the aortic branches that lead to abdominal organs are lateral branches.
- D. The most common location for an aortic dissection is at the iliac bifurcation.

105. Which of the following is a condition that cannot be evaluated with Transcranial Doppler?

- A. MCA disease
- B. temporal arteritis
- C. vertebro-basilar disease
- D. patent foramen ovale

106. Which of the following is an abnormal flow change caused by congestive heart failure?

- A. increased velocities in the bilateral common carotid arteries
- B. decreased ankle pressures
- C. pulsatile flow in the proximal hepatic veins
- D. monophasic portal flow

107. A patient presents for a venous insufficiency exam. The exam demonstrates normal bilateral great and small saphenous veins. The deep system is not identified well in either leg. Both calves have multiple areas of small, clustered veins superficially. What should you review in the patient chart to explain these findings?

- A. medical history for Klippel Trenaunay Webber syndrome
- B. treatment history for RF ablation of the bilateral GSVs
- C. surgical history for venous ligation of the bilateral GSVs
- D. medical history for Paget Schroetter syndrome

108. Which of the following disorders would demonstrate normal digital pressures in the upper extremity?

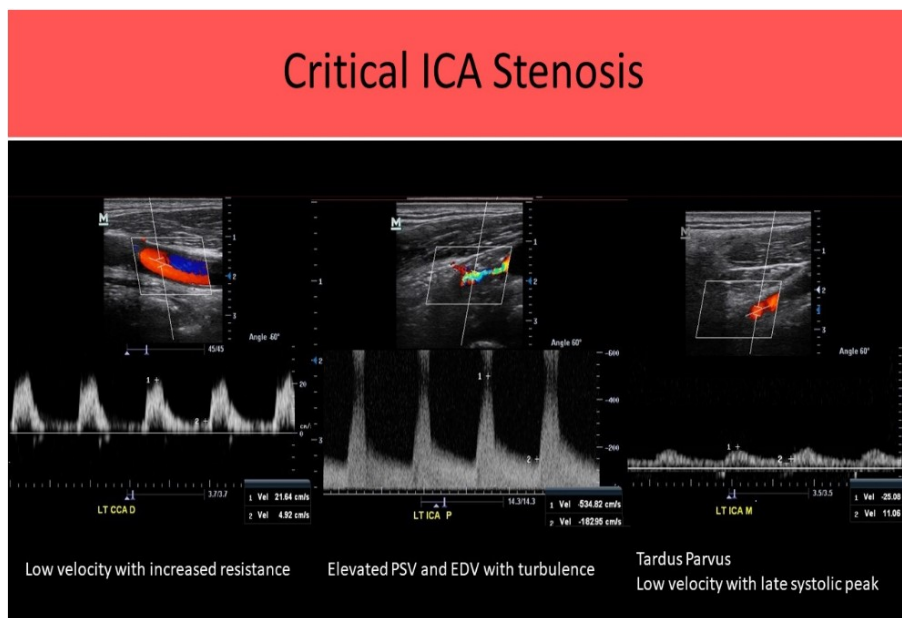
- A. Raynaud disease
- B. Thoracic outlet syndrome
- C. Buerger disease
- D. SVC syndrome

109. _____ is the "gold standard" exam technique for diagnosis of venous disease in the lower extremities.

- A. Arteriography
- B. Venography
- C. Doppler Ultrasound
- D. MR Angiography (MRA)

110. Which of the following would be used to describe a Doppler waveform taken in the mid ICA with a critical stenosis of the proximal segment?

- A. increased pulsatility index
- B. peaked
- C. damped
- D. increased resistive index



111. If both carotid waveforms look like the image above, you should check the patient chart to see if they have: view image

- A. constrictive pericarditis

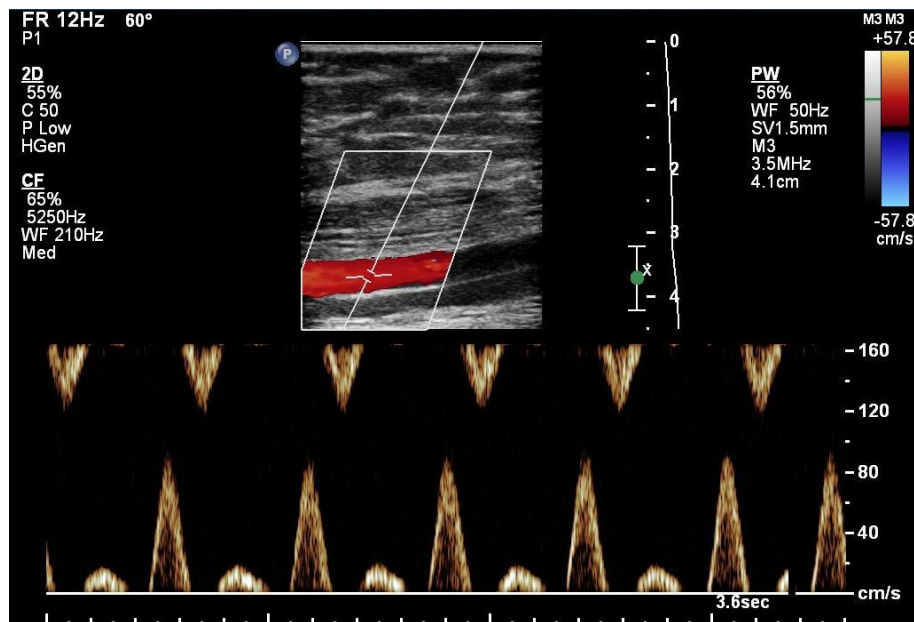
- B. an intra-aortic balloon pump
- C. severe aortic stenosis
- D. subclavian steal

112. Which of the following screening techniques for peripheral arterial disease can distinguish between a stenosis and an occlusion in a vessel?

- A. Segmental pressure evaluations
- B. Doppler evaluations
- C. Pulse Volume Recording
- D. Photoplethysmography

113. What is the best way to adjust the console controls to eliminate the Doppler artifact?

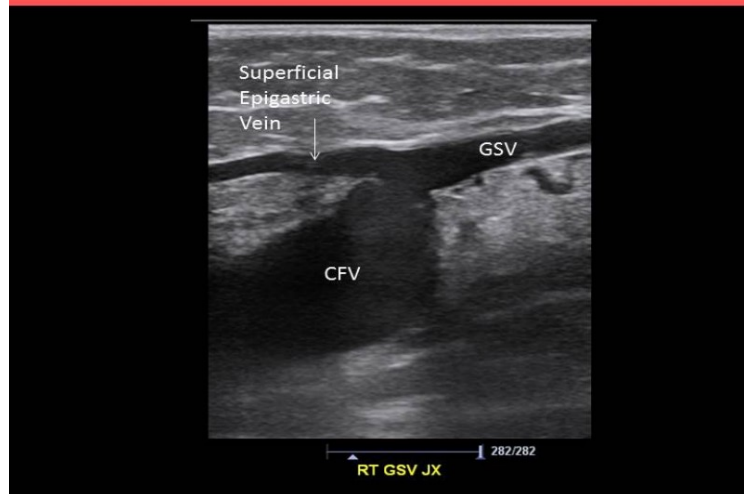
- A. increase the sweep speed
- B. turn down the Doppler gain
- C. raise the baseline
- D. increase the PW Doppler scale



114. Which of the following locations has a direct connection between the deep and superficial veins of the leg?

- A. adductor canal
- B. groin
- C. hunter canal
- D. ankle

Saphenofemoral Junction (SFJ)



115. Which of the following exams does not use PW or CW Doppler?

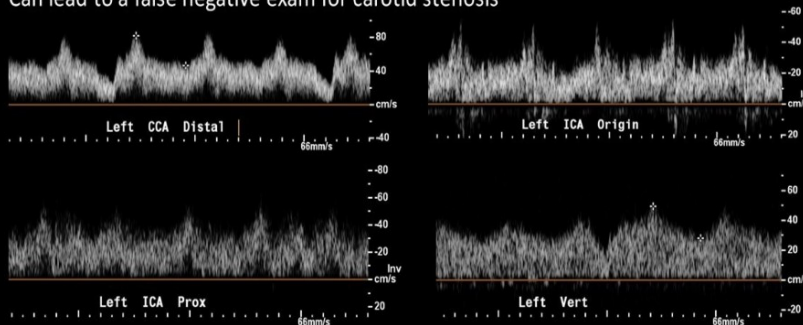
- A. transcranial exam
- B. pulse volume recording
- C. periorbital exam
- D. extracranial exam

116. A patient presents for a carotid ultrasound with a history of myocardial infarction and a left ventricular assist device is present. Which of the following describes the expected flow pattern in the normal common carotid artery?

- A. low-velocity flow with delayed upstroke and loss of pulsatility
- B. high-velocity flow with accelerated upstroke
- C. antegrade flow in systole and retrograde flow in diastole
- D. triphasic waveform with increased systolic velocity

LV Assist Device – Carotid Doppler

Diminished peak systolic velocity, increased diastolic flow
 Mean flow velocity usually remains unchanged with changes to pump settings
 Can lead to a false negative exam for carotid stenosis



Courtesy of Cleveland Clinic

117. Which of the following symptoms would be most suggestive of significant disease in the middle cerebral artery(MCA)?

- A. Ipsilateral leg paresis
- B. Contralateral leg paresis
- C. Contralateral face and arm paresis
- D. Ipsilateral face and arm paresis

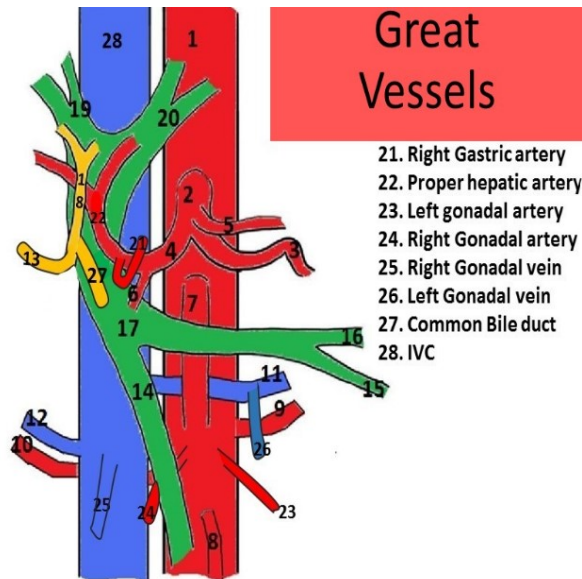
118. Lymphedema is commonly related to:

- A. Leriche syndrome
- B. peripheral arterial disease
- C. a pulmonary embolism
- D. a history of radiation therapy or cancer

119. Which abdominal vessel arises from the anterior surface of the aorta approximately 2 cm distal to the celiac axis?

- A. gastroduodenal artery
- B. inferior mesenteric artery
- C. superior mesenteric artery
- D. common hepatic artery

1. Aorta
2. Celiac
3. Splenic artery
4. Common hepatic artery
5. Left Gastric artery
6. Gastroduodenal artery
7. SMA
8. IMA
9. LRA
10. RRA
11. LRV
12. RRV
13. Cystic duct
14. SMV
15. IMV
16. Splenic Vein
17. Portal confluence
18. Common Hepatic duct
19. RPV
20. LPV



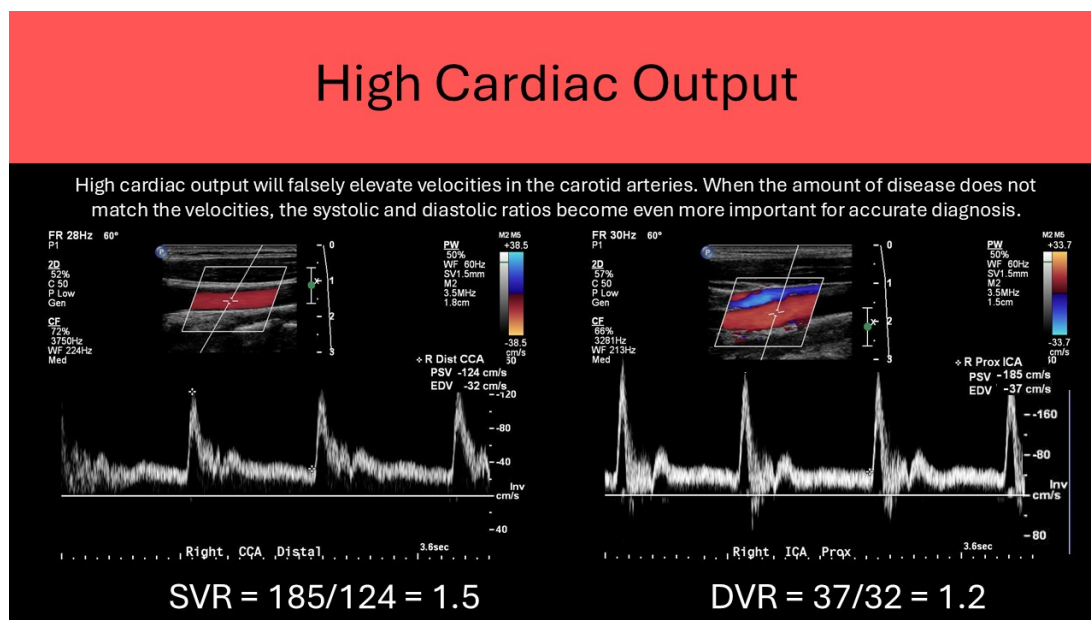
120. The small saphenous vein:

- A. originates anterior to the lateral malleolus and ascends along the midline aspect of posterior calf
- B. originates posterior to the medial malleolus and ascends along the midline aspect of posterior calf
- C. originates posterior to the lateral malleolus and ascends along the midline aspect of posterior calf
- D. originates immediately posterior to the Achilles' tendon attachment at the heel and ascends along the midline aspect of posterior calf

121. You perform a carotid Doppler exam that demonstrates minimal atheroma formation bilaterally. You record waveforms that demonstrate laminar flow with the following velocities: Right Proximal CCA PSV 145cm/s, EDV 35cm/s Right Proximal ICA PSV 185cm/s, EDV 50cm/s Left Proximal CCA PSV 140cm/s, EDV 38cm/s Left Proximal ICA PSV 190cm/s, EDV 40cm/s What will you report on your worksheet?

- A. Flow velocities are increased bilaterally, which is most likely related to increased cardiac output

- B. Flow velocities are increased bilaterally, which is most likely related to stenosis at the aortic valve
- C. Flow velocities are normal bilaterally
- D. Flow velocities are increased bilaterally, which is most likely related to soft plaque formation not visible on 2D and color evaluation



122. Which of the following is related to the appearance of Rouleaux formation on the image?

- A. Thoracic outlet syndrome
- B. Raynaud Syndrome
- C. Intracranial collateral flow
- D. DVT

123. You are performing a left lower extremity venous duplex exam. DVT is identified in the popliteal and femoral veins. Which of the following must also be included in the exam protocol for this patient?

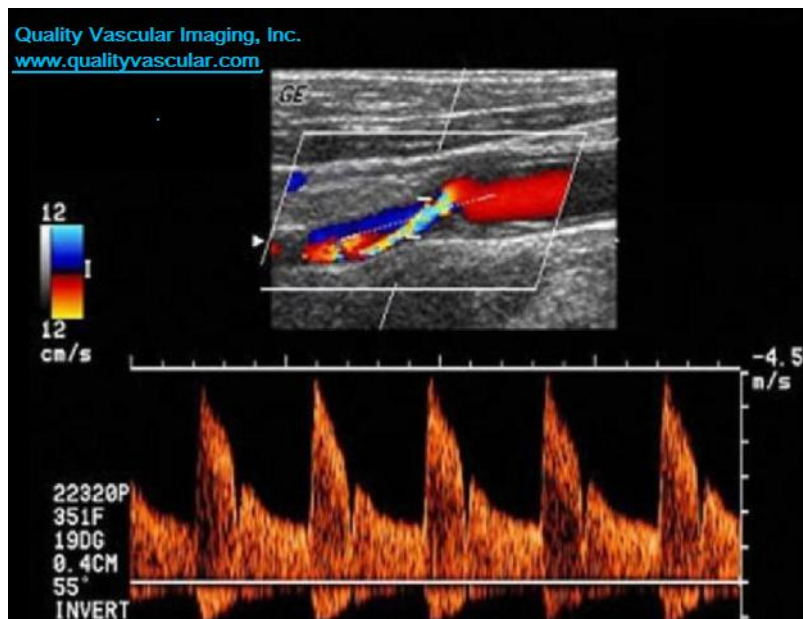
- A. arterial flow pattern in the arteries associated with the thrombosed veins
- B. pulse volume recording of the calf and thigh of the contralateral leg
- C. flow in the contralateral common femoral vein
- D. ankle brachial index

124. Which of the following describes collateral flow from the ECA to the ICA on the same side of the body?

- A. lingual artery to the MCA
- B. posterior communicator to the basilar artery
- C. superior thyroidal artery to the ACA
- D. superficial temporal artery to the ophthalmic artery

125. The PW Doppler tracing displays what level of ICA stenosis?

- A. 0-49%
- B. 50-69%
- C. 70-99%
- D. 100%



126. The most common location of a Baker cyst is:

- A. Inguinal canal
- B. Medial popliteal fossa
- C. Medial antecubital fossa
- D. Anterior patellar space

127. Which of the following normal vessels will collapse when you ask the patient to sniff quickly with moderate exertion?

- A. common femoral vein
- B. basilic vein
- C. subclavian vein
- D. cephalic vein

128. The weight of the column of blood within the veins when standing is called?

- A. Systemic Pressure
- B. Hydrostatic Pressure
- C. Pulmonary Pressure
- D. Transmural Pressure

129. What is the most common cause of upper extremity DVT in the normal ambulatory patient population?

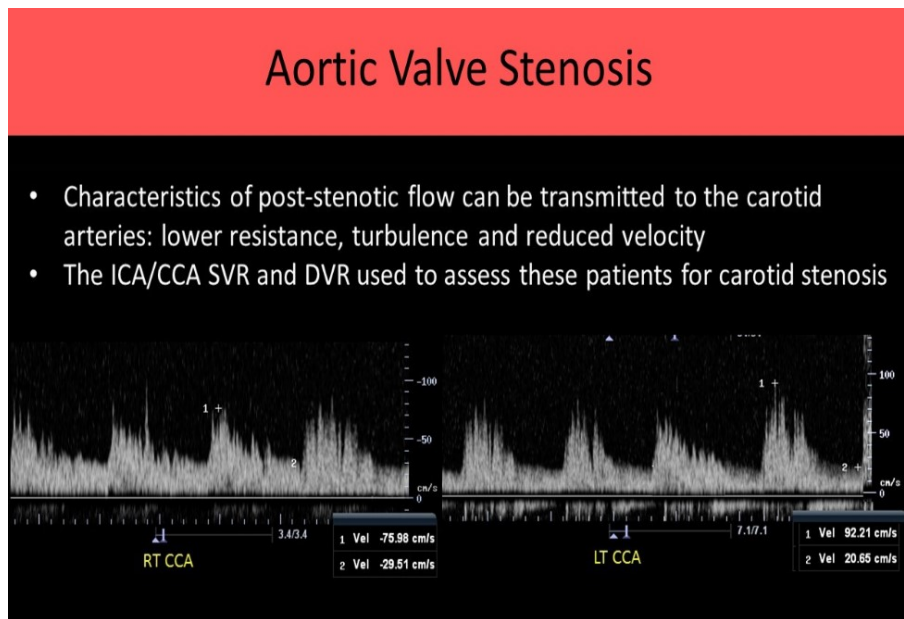
- A. shoulder surgery
- B. Paget - Schroetter Syndrome
- C. radiation therapy treatments to the chest
- D. indwelling catheters

130. What patient position should be used when mapping the great saphenous vein for harvest?

- A. left lateral decubitus
- B. reverse Trendelenburg
- C. Trendelenburg
- D. supine

131. Which of the following would lead to underestimation of stenosis in the internal carotid artery?

- A. Significant aortic valve stenosis
- B. Elevated heart rate
- C. Short area of focal stenosis
- D. Compensatory flow due to stenosis on other side



132. Put the following terms in order from largest to smallest in size. capillary, aorta, femoral artery, arteriole, tibial artery

- A. capillary, aorta, femoral artery, tibial artery, arteriole
- B. aorta, capillary, femoral artery, tibial artery, arteriole
- C. aorta, femoral artery, tibial artery, arteriole, capillary
- D. capillary, arteriole, tibial artery, femoral artery, aorta

133. Where is the arc of Rioloan found?

- A. within the brain
- B. at the popliteal fossa
- C. within the mesentery
- D. within the kidney

134. A patient presents for a carotid ultrasound due to recent episodes of near syncope. Noticeable respiratory variation in the peak velocities of the carotid arteries is present. What cardiac complication should be suspected?

- A. tricuspid stenosis
- B. low cardiac output
- C. mitral stenosis
- D. cardiac tamponade

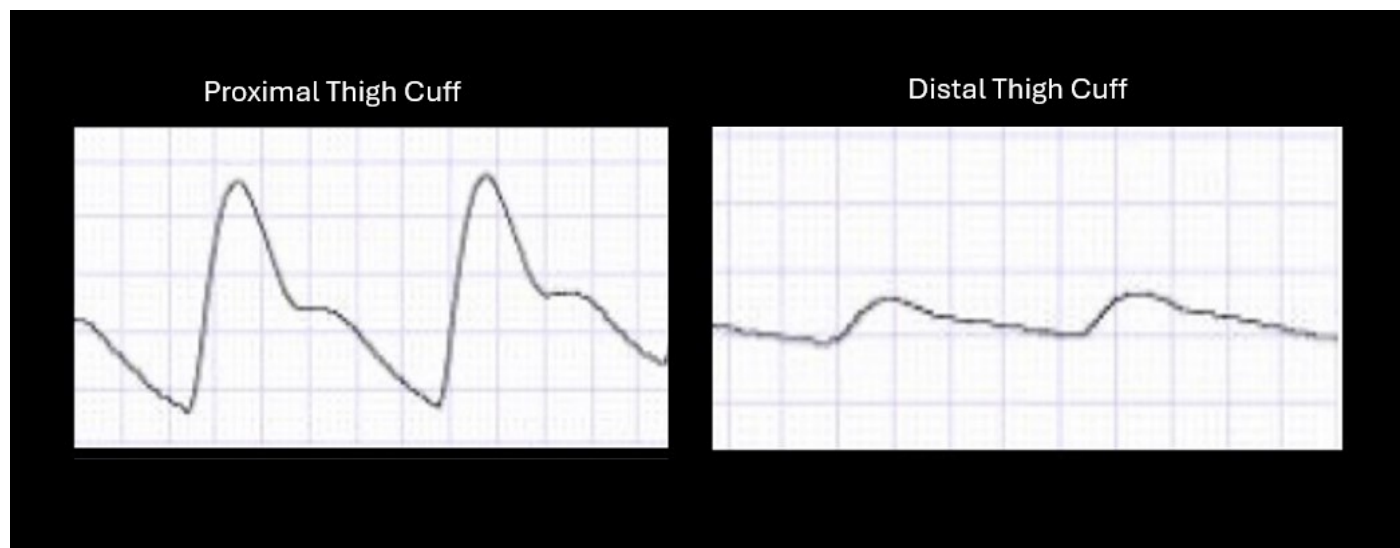
135. A patient presents for an exercise arterial exam after an abnormal resting exam. The report from the resting exam describes: elevated velocities that indicate a stenosis of >50% in the distal femoral artery on the left leg, biphasic waveforms in the PTA and DPA and an ABI of 0.96. There is no change in the waveforms or decrease in ankle pressures in the left leg with exercise. What could explain these findings?

- A. Raynaud syndrome
- B. congestive heart failure

- C. obstructed venous outflow
- D. extensive collateral formation

136. The PVR tracings from a lower extremity arterial exam. Which of the following explains these findings?

- A. stenosis distal to the proximal thigh cuff
- B. stenosis proximal to the proximal thigh cuff
- C. unable to determine with PVR tracing from the calf
- D. stenosis distal to the distal thigh cuff



137. If the image displayed is a transverse view of the posterior right lower leg, Letter B correspond to:

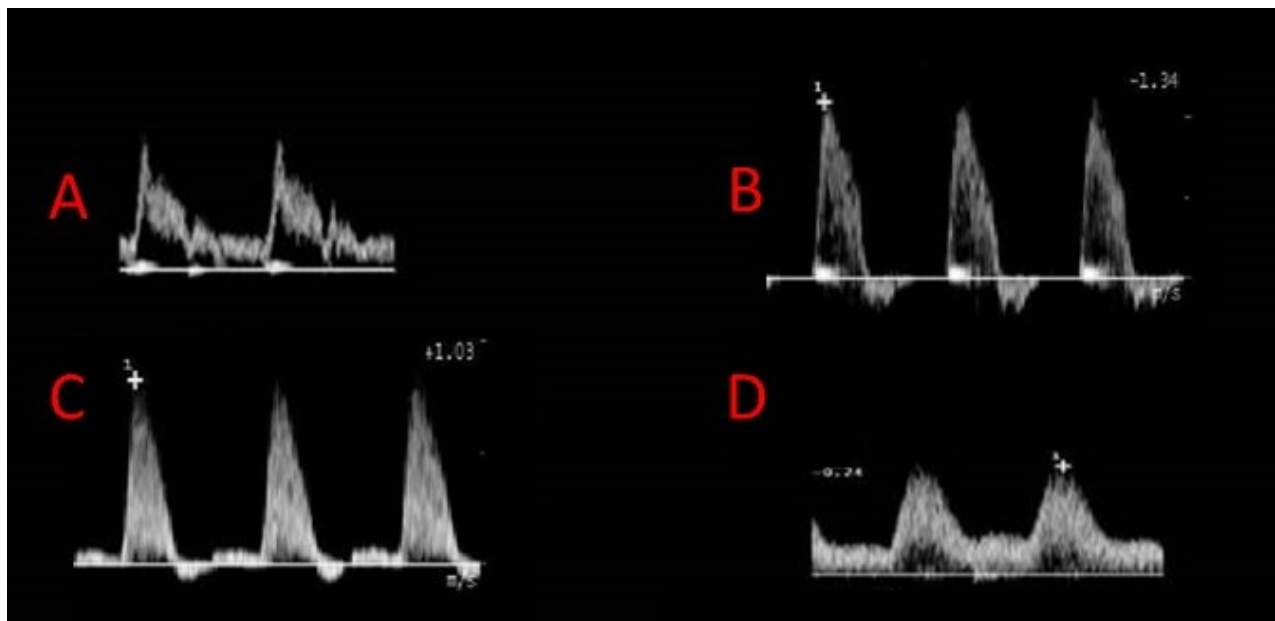
- A. medial gastrocnemius veins
- B. peroneal veins
- C. posterior tibial veins
- D. lateral gastrocnemius veins

138. When evaluating a pregnant patient for DVT, the patient should be placed in the _____ position.

- A. supine
- B. right oblique or decubitus
- C. Trendelenburg
- D. left oblique or decubitus

139. Which waveform demonstrates the lowest resistance?

- A. B
- B. C
- C. A
- D. D



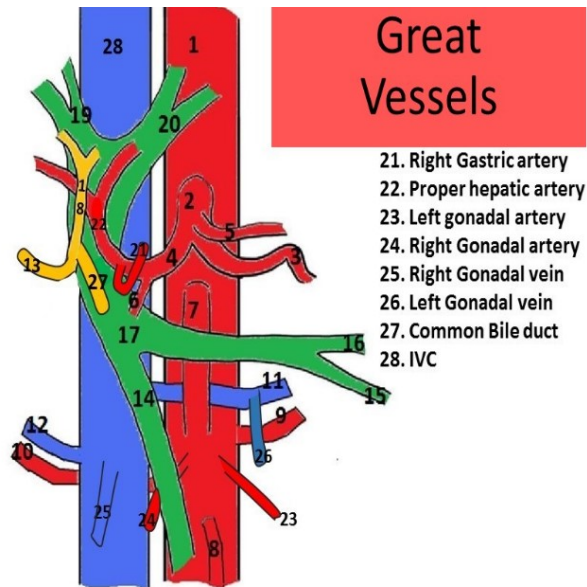
140. You identify acute thrombosis of the left popliteal and femoral vein, and the proximal end of the clot has a free-floating tail. What should you do next?

- A. Check the patient chart for a history of Coumadin therapy
- B. Call for a code blue due to the significant increased risk of embolism
- C. Tell the patient to limit all movement because the DVT is unresolved and you need to consult the reading physician immediately
- D. Check the great saphenous vein for thrombus propagation

141. Which of the following is true regarding scrotal varicoceles?

- A. usually occurs on left side
- B. strong association with history of cryptorchidism
- C. veins will reduce in size with the Valsalva maneuver
- D. usually occurs bilaterally

1. Aorta
2. Celiac
3. Splenic artery
4. Common hepatic artery
5. Left Gastric artery
6. Gastroduodenal artery
7. SMA
8. IMA
9. LRA
10. RRA
11. LRV
12. RRV
13. Cystic duct
14. SMV
15. IMV
16. Splenic Vein
17. Portal confluence
18. Common Hepatic duct
19. RPV
20. LPV



Great Vessels

21. Right Gastric artery
22. Proper hepatic artery
23. Left gonadal artery
24. Right Gonadal artery
25. Right Gonadal vein
26. Left Gonadal vein
27. Common Bile duct
28. IVC

142. Poiseuille's Law states that the _____ has the most significant effect on the blood flow in a vessel.

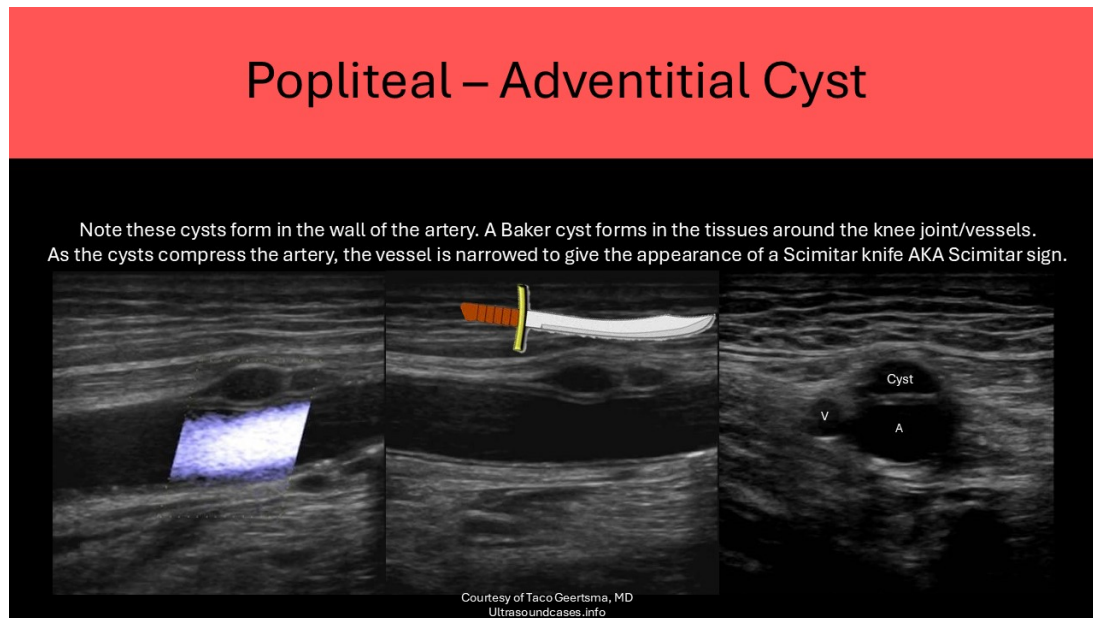
- A. pressure gradient
- B. vessel length
- C. vessel radius
- D. fluid viscosity

143. A recent CT scan demonstrates an acute infarct in the parietal lobe and a new blockage in the proximal left middle cerebral artery. Which of the following is the most likely source for the embolism?

- A. Thrombus formation in the left jugular vein
- B. Plaque broke away from the wall of the proximal external cerebral artery
- C. Plaque broke away from the wall of the left anterior cerebral artery
- D. Plaque broke away from the sclerotic aortic valve

144. The Scimitar sign is associated with:

- A. Femoral to Femoral bypass graft
- B. Median arcuate ligament syndrome
- C. Bovine arch
- D. Adventitial cysts



145. When Thrombin is injected into a pseudoaneurysm, the needle puncture site should be:

- A. on the lateral aspect of the groin
- B. within 1cm of the neck
- C. within 5mm of the neck
- D. located as far from the neck as possible

146. If greater saphenous venous flow velocity is approximately 20cm/s and appears continuous, which other vessel should be evaluated for a cause for the flow abnormality?

- A. internal iliac vein
- B. short saphenous vein
- C. profunda vein
- D. femoral and popliteal vein

147. A 9-year-old presents with chronic fatigue and decreased palpable femoral pulses. The ankle brachial index is performed with the following results. Lt arm 168mmHg Lt ankle 95mmHg Rt arm 174mmHg Rt ankle 86mmHg Which of the following is the most likely explanation of the findings?

- A. There is most likely a coarctation of the descending aorta.
- B. There is most likely atherosclerotic vascular disease present in both lower extremities.
- C. There is most likely a subclavian steal present.
- D. The bilateral ABIs are normal.

148. Rest pain is associated with ankle pressures less than _____ and toe pressures less than _____ will be associated with a non-healing wound.

- A. 100mmHg, 30mmHg
- B. 70mmHg, 50mmHg
- C. 50mmHg, 30mmHg
- D. 50mmHg, 50mmHg

149. A patient presents with a script for an upper extremity PPG exam to rule out Raynaud syndrome. How will you perform the exam?

- A. Place PPG sensors on the tips of the fingers before and after cold sensitivity testing.
- B. Place the PPG sensors at the antecubital fossa and apply tourniquets at multiple levels on the arm to assess changes in flow.
- C. Place PPG sensors on the wrists before and after cold sensitivity testing.
- D. Raynaud syndrome cannot be evaluated with PPG techniques.

150. You are evaluating a patient with suspected Raynaud disease using PPG. After obtaining a baseline tracing, you place the hand in cold water for 2 minutes. Once the hand is removed, PPG sensors are reapplied to the fingertips. Which of the following correctly describes normal results for this exam?

- A. waveforms return to baseline within 7 minutes after removing the hand from the cold water
- B. waveform amplitude increases to at least 2X the original
- C. waveform amplitude increases to at least 4X the original
- D. waveforms return to baseline within 5 minutes after removing the hand from the cold water

151. Which of the following transducers would be preferred for a lower extremity saphenous mapping on an average sized patient?

- A. 7-10MHz sector
- B. 7-10MHz linear
- C. 5-7MHz curvilinear
- D. 5-7MHz linear

152. It is important to wait at least one minute before repeating what measurement?

- A. brachial pressure on an arm with a hemodialysis graft
- B. ankle blood pressure
- C. MCA mean velocity
- D. volume flow in a hemodialysis graft

153. Hyperostosis is a limitation of which exam?

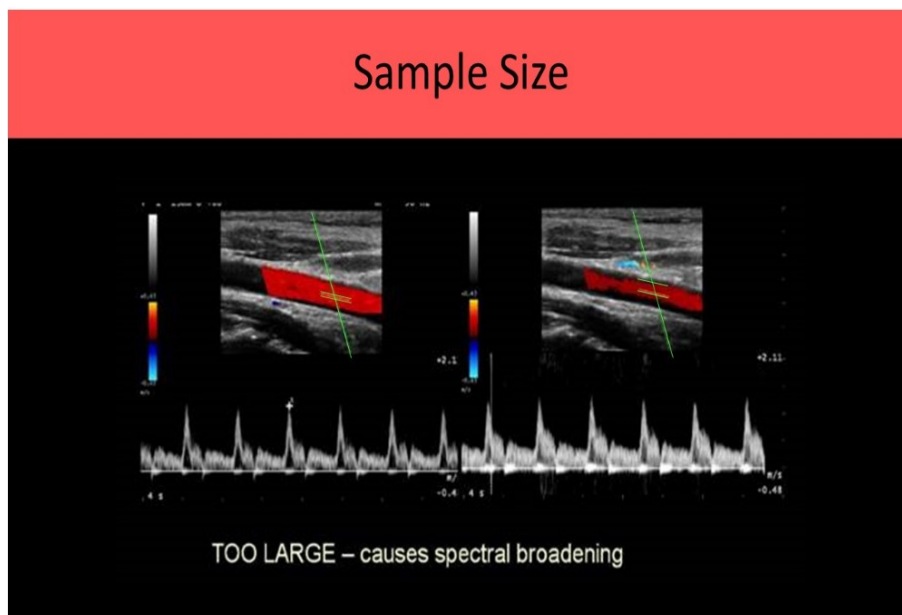
- A. lower extremity arterial Doppler exam
- B. lower extremity venous Doppler exam
- C. transcranial Doppler exam
- D. renal artery Doppler exam

154. When evaluating heavily calcified arteries, what system settings should be increased to improve the demonstration of flow in the vessels?

- A. color gain and angle of insonation
- B. persistence and sensitivity
- C. velocity scale and persistence
- D. frequency and output power

155. Which of the following system adjustments would increase the size of the spectral window?

- A. changing from a 4mm to 2mm sample size
- B. tortuous vessel
- C. decreasing rejection
- D. changing from a 2mm to 4mm sample size



156. In photoplethysmography, the sensor uses _____ to detect _____ changes.

- A. pressure cu■, blood pressure
- B. light, blood pressure
- C. light, blood volume
- D. thin mercury tubing, volume

157. A segmental pressure exam demonstrates the following pressures: right arm: 142mmHg right leg: proximal thigh 140mmHg distal thigh 126mmHg calf 116mmHg ankle 112mmHg. Which of the following could explain these results?

- A. infrapopliteal disease
- B. pelvic mass
- C. pelvic mass or mid femoral artery stenosis
- D. medial calcification

158. A segmental pressure exam demonstrates the following pressures: distal thigh 144, calf 166, ankle 132. Which of the following could explain these results?

- A. medial calcification or venous insufficiency
- B. improper cu■ placement and incorrect cu■ size, but not medial calcification
- C. incorrect cu■ size would be the only possible cause

D. incorrect cu■ size, medial calcification and improper cu■ placement

159. Which of the following is a reason to perform exercise with a segmental pressure exam?

- A. to evaluate volume changes in the limb
- B. to evaluate the effects of vasoconstriction
- C. to differentiate an SFA stenosis from an occlusion
- D. to differentiate true claudication from pseudoclaudication

160. Increasing the spectral Doppler sample size will increase the _____.

- A. sweep speed
- B. size of the spectral window
- C. spectral broadening
- D. Doppler shift

161. What is indicated by the red arrows on this subcostal image?

- A. Abdominal Aorta
- B. Mirror image artifact
- C. Portal Vein
- D. IVC



162. Which of the following is not a patient demographic that is collected to evaluate patient population statistics for those with vascular disease?

- A. Employment history
- B. Ethnicity
- C. Gender
- D. Age

163. Which of the following requires a Doppler cursor angle of 60 degrees or less for accurate reporting of numerical values?

- A. pulsatility index
- B. resistive index
- C. peak systolic velocity
- D. S/D ratio

164. Which of the following can be used to eliminate aliasing seen on a CW Doppler tracing?

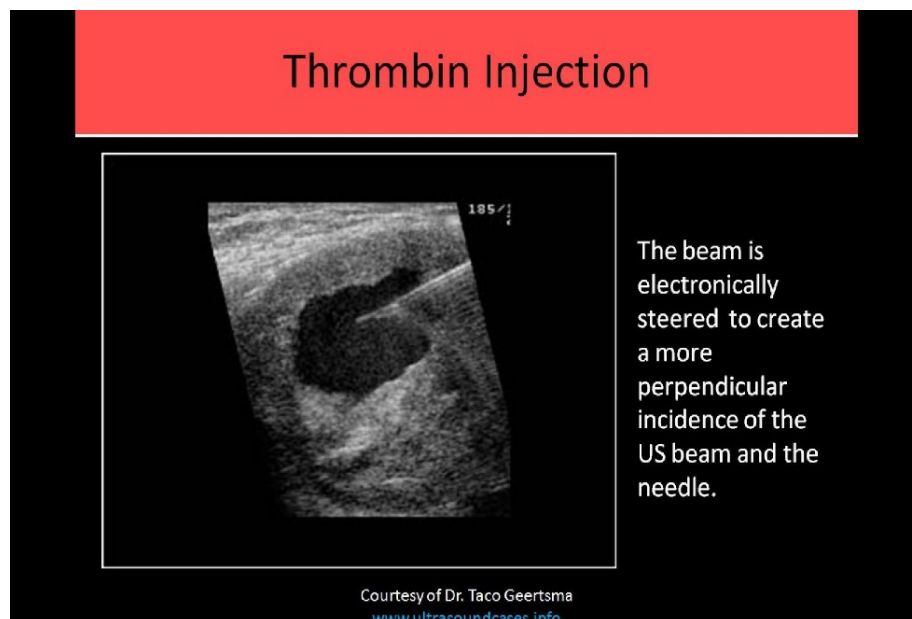
- A. decrease probe frequency
- B. increase the scale
- C. adjust baseline, increase scale and decrease probe frequency
- D. none of the above

165. Which of the following statements is correct regarding intraoperative duplex evaluation during a carotid endarterectomy?

- A. intraoperative duplex evaluation cannot be performed during a carotid endarterectomy.
- B. The technique is used to assess the immediate flow changes within the vessel undergoing the endarterectomy.
- C. Due to the increased attenuation of sound caused by air in the extracranial vessels, the lowest probe frequency available is preferred for the exam.
- D. Sterile gel is required because a sterile sheath is not used due to potential acoustic artifacts.

166. If the physician is having trouble visualizing the needle on the ultrasound screen during a thrombin injection, how can you help to improve the visualization of the needle?

- A. increase the output power to the maximum setting
- B. use electronic steering to increase the incident angle between the US beam and the long axis needle to a value closer to 90 degrees
- C. use electronic steering to reduce the incident angle between the US beam and the long axis needle to a value closer to 0 degrees
- D. use a lower frequency probe

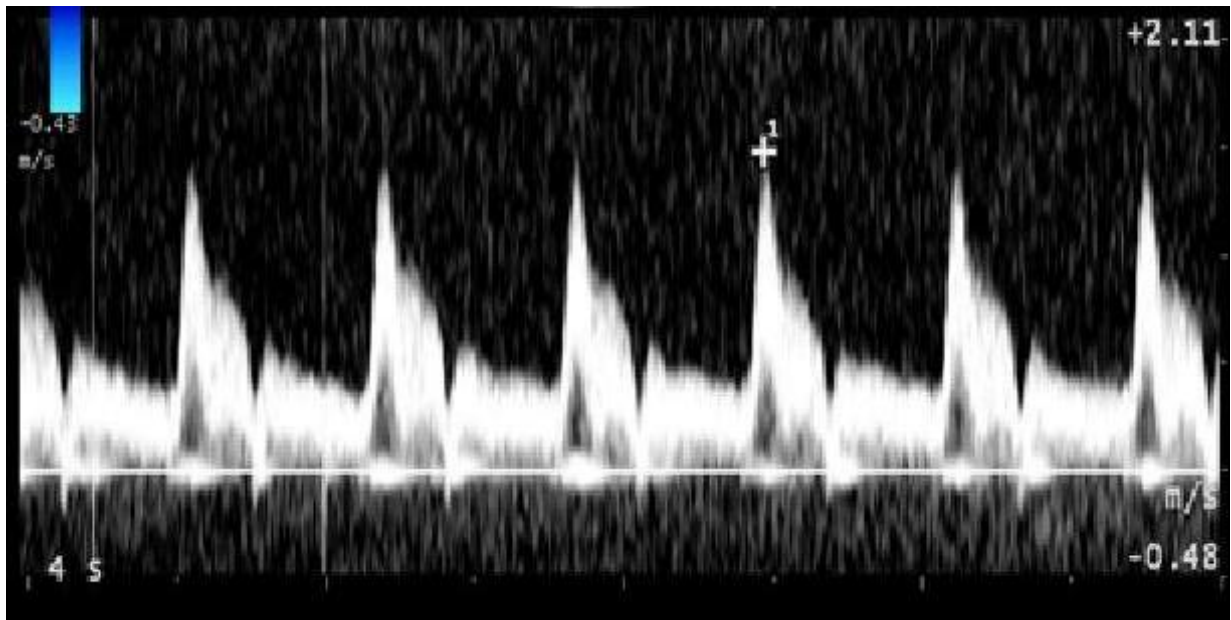


167. What change in system controls can reduce the appearance of color displayed over vessel walls?

- A. decreasing color PRF
- B. decreasing color priority
- C. increasing color threshold
- D. increasing persistence

168. What can be done to reduce the spectral broadening on the image?

- A. decrease rejection
- B. decrease Doppler Gain
- C. decrease the PRF
- D. decrease wall filter



169. When preparing the sterile tray for an invasive procedure, the cover on the tray should be opened:

- A. while wearing sterile gloves
- B. by the radiologist
- C. no more than 2 hours before the procedure begins
- D. by pulling the far corner of the wrapper toward you

170. If the great saphenous vein is being used to create an in-situ vein graft,

- A. the graft diameter will be greatest at the distal end of the graft
- B. all perforator veins must be marked on the vein mapping so they can be closed
- C. the valves will remain in place and provide little resistance to arterial flow
- D. all perforator veins must be evaluated for patency because the graft will not function properly with obstructed perforator veins

Answer Key

Questions 34 and 78 are image-click items in the source file, so they are shown with "-" in the key.

1	A	2	A	3	D	4	C
5	B	6	A	7	B	8	D
9	A	10	B	11	D	12	C
13	A	14	A	15	B	16	C
17	A	18	C	19	C	20	D
21	A	22	C	23	C	24	A
25	D	26	A	27	B	28	C
29	D	30	B	31	C	32	D
33	C	34	-	35	A	36	A
37	D	38	D	39	B	40	D
41	C	42	B	43	C	44	C
45	C	46	A	47	D	48	A
49	B	50	D	51	C	52	C
53	D	54	C	55	A	56	B
57	A	58	D	59	B	60	D
61	D	62	D	63	C	64	C
65	B	66	A	67	D	68	B
69	A	70	A	71	D	72	B
73	B	74	B	75	C	76	C
77	C	78	-	79	D	80	D
81	D	82	A	83	D	84	B
85	D	86	A	87	C	88	B
89	B	90	D	91	B	92	B
93	C	94	B	95	C	96	D
97	B	98	A	99	A	100	B
101	C	102	D	103	B	104	A
105	B	106	B	107	A	108	D
109	C	110	C	111	B	112	B
113	C	114	B	115	B	116	A
117	C	118	D	119	C	120	C
121	A	122	D	123	C	124	D
125	C	126	B	127	C	128	B
129	B	130	B	131	A	132	C
133	C	134	D	135	D	136	A
137	D	138	D	139	D	140	A
141	A	142	C	143	D	144	D
145	D	146	D	147	A	148	C

149	A	150	D	151	B	152	B
153	C	154	B	155	A	156	C
157	B	158	D	159	D	160	C
161	B	162	A	163	C	164	D
165	B	166	B	167	B	168	B
169	D	170	B				